QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

rs-soroban-env @ 0c99950e9612 · Stellar Development Foundation — public repository (passive review)

Organization	QSI
Report date	2026-11-06 16:05 UTC
Report version	1
Audit run	b6dad59b-c7b1-4e8f-86d3-2c4d7916c720 · Document audit · 2026-11-06 16:05 UTC
Analyzer engine	qsi-engine 1.3.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`9b85f3462552f82353205b89d9274711efee1e99748e272a6e7abd94e0172d1e`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

18	0	0	10	6
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	9
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	7
Improvement recommendation	2
Insufficient data	0

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-19557	Medium	Quantum-vulnerable Ed25519: transaction and message signing (2 file(s)) preliminary	35.6
QSI-2026-19561	Medium	Quantum-vulnerable ECDSA: transaction and message signing (2 file(s)) preliminary	35.6
QSI-2026-19558	Medium	Quantum-vulnerable Ed25519: role not determined (9 file(s)) preliminary	34.2
QSI-2026-19559	Medium	Quantum-vulnerable BLS12-381: role not determined (5 file(s)) preliminary	34.2
QSI-2026-19560	Medium	Quantum-vulnerable BN254 (alt_bn128): role not determined (2 file(s)) preliminary	34.2

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Post-quantum maturity: Level 2 of 4 — post-quantum in use outside the critical paths. Next step: Extend post-quantum protection to the critical roles that stay classical, starting with what can be recorded today (wrapped keys, channel key exchange).

Scope and boundaries

Mode: Document audit

Repository: https://github.com/stellar/rs-soroban-env @ 0c99950e96128beba3319aa96b116646ce974e69 — evidence links open the lines at this commit.

Supplied inputs

256 files, 2036998 bytes in total, 7 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.3.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
dependencies	checked	52	Cargo.lock: 273 dependencies parsed (273 pinned); Cargo.toml: 8 dependencies parsed (8 pinned); soroban-bench-utils/Cargo.toml: 4 dependencies parsed (2 pinned)
source_code	checked	202	soroban-bench-utils/src/lib.rs: 3 lines scanned with 16 rule patterns and 267 registry patterns; soroban-bench-utils/src/rusagev4.rs: 62 lines scanned with 16 rule patterns and 267 registry patterns; soroban-bench-utils/src/tracker.rs: 193 lines scanned with 16 rule patterns and 267 registry patterns
secrets	checked	1	soroban-env-host/src/host/invocation_metering.rs: 7 redacted value(s) not reported (comments, test modules, fixture declarations, placeholders, values the source annotates as published, or hex values not bound to a secret name)
entropy	checked	1	random-number generation: 202 code files read with 0 macro definitions from the tree; 0 ineffective guard(s), 0 weak random-bytes provider(s), 0 weak key-generation site(s)
osv_advisories	not checked	1	3 package(s) are built from git repositories (forks or unreleased code), so registry advisories were not looked up for them: soroban-wasmi, wasmi_arena, wasmi_core
osv_advisories	checked	1	340 pinned packages queried at OSV.dev; 11 affected

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 256 (input files: C)

Not checked

- osv_advisories: 3 package(s) are built from git repositories (forks or unreleased code), so registry advisories were not looked up for them: soroban-wasmi, wasmi_arena, wasmi_core

Architecture overview

Static review of the source tree rs-soroban-env at 0c99950e96128beba3319aa96b116646ce974e69. 256 files analysed; of 1132 files, 174 are test, fuzzing, benchmark or example code and were left out, 702 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. Files no analyzer reads include: .devcontainer/Dockerfile, .devcontainer/devcontainer.json, .github/CODEOWNERS, .github/pull_request_template.md, .github/workflows/bump-version.yml. 31 files in build-output or dependency-cache directories (.git, .hg, .venv, __pycache__, build, dist, node_modules, target, vendor, venv) were not read. No requests were sent to the owner's systems.

Overall security posture

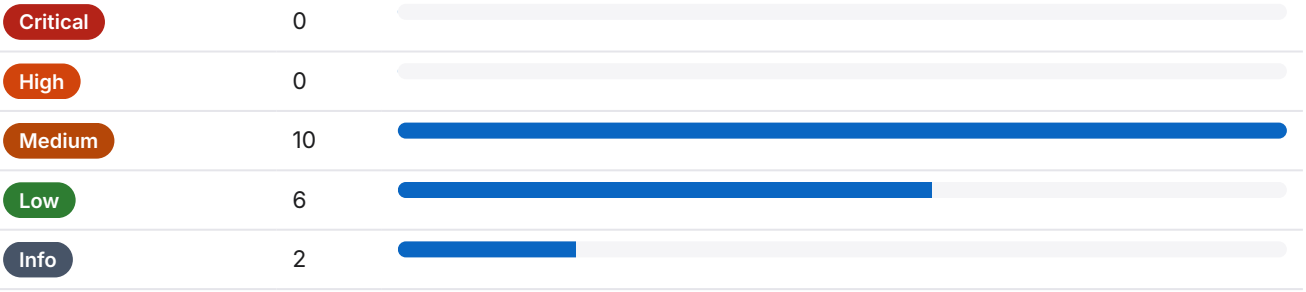
80
Indicative posture index (0–100)

50.0%
Quantum-vulnerable share of inventory

76
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-19557	Quantum-vulnerable Ed25519: transaction and message signing (2 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19561	Quantum-vulnerable ECDSA: transaction and message signing (2 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19558	Quantum-vulnerable Ed25519: role not determined (9 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19559	Quantum-vulnerable BLS12-381: role not determined (5 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19560	Quantum-vulnerable BN254 (alt_bn128): role not determined (2 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19562	Quantum-vulnerable ECDSA: role not determined (6 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19563	Quantum-vulnerable Ed25519: authentication and identity (certificates, tokens, device attestation) (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19569	serde_with 3.12.0 is affected by GHSA-7gcf-g7xr-8hxj	Known vulnerable component	Medium	Low
QSI-2026-19574	serde_with 3.14.0 is affected by GHSA-7gcf-g7xr-8hxj	Known vulnerable component	Medium	Low
QSI-2026-19570	time 0.3.41 is affected by GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009)	Known vulnerable component	Medium	Low, preliminary
QSI-2026-19568	rand 0.8.5 is affected by GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097)	Known vulnerable component	Low	Medium
QSI-2026-19571	tracing-subscriber 0.3.18 is affected by GHSA-xwfj-jgwm-7wp5 (CVE-2025-58160, RUSTSEC-2025-0055)	Known vulnerable component	Low	Low
QSI-2026-19566	keccak 0.1.4 is affected by GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012)	Known vulnerable component	Low	Low, preliminary
QSI-2026-19564	crypto-bigint 0.5.2 is affected by RUSTSEC-2026-0387 (no severity rating)	Known vulnerable component	Low	Low, preliminary

QSI-2026-19565	ctutils 0.4.2 is affected by RUSTSEC-2026-0385 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-19572	crypto-bigint 0.5.4 is affected by RUSTSEC-2026-0387 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-19567	paste 1.0.12: informational advisory RUSTSEC-2024-0436 (unmaintained)	Improvement recommendation	Info	Low
QSI-2026-19573	paste 1.0.14: informational advisory RUSTSEC-2024-0436 (unmaintained)	Improvement recommendation	Info	Low

Detailed findings

QSI-2026-19557 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: transaction and message signing (2 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

4 reference(s) to Ed25519 across 2 file(s), e.g. soroban-env-host/src/builtin_contracts/stellar_asset_contract/metadata.rs:158. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Ed25519:transaction_signing
Data source	soroban-env-host/src/builtin_contracts/stellar_asset_contract/metadata.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:05 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open soroban-env-host/src/builtin_contracts/stellar_asset_contract/metadata.rs:158 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · soroban-env-host/src/builtin_contracts/stellar_asset_contract/metadata.rs · line 158 · SHA-256 3d7fb8af17a89202… · File 12ca5219bd19e98b…

```
157 |     s.push(':');
158 |     s.push_str(&ed25519::PublicKey(issuer.to_array()?).to_string());
159 |     let name = String::try_from_val(e, &e.string_new_from_slice(s.as_bytes())?)?;
```

file_excerpt · soroban-env-host/src/testutils.rs · line 15 · SHA-256 c8d71454e9e88769… · File b39bf85a85009499…

```
 14 | };
 15 | use ed25519_dalek::SigningKey;
 16 | use rand::RngCore;
```

file_excerpt · soroban-env-host/src/testutils.rs · line 316 · SHA-256 ccca624ff7d49ed3… · File b39bf85a85009499…

```
315 |         n_signers: usize,
316 |     ) -> (Host, Vec<AddressObject>, Vec<ed25519_dalek::SigningKey>) {
317 |         use crate::builtin_contracts::testutils::{
```

file_excerpt · soroban-env-host/src/testutils.rs · line 376 · SHA-256 5c948ca4d4acf0e6… · File b39bf85a85009499…

```
375 |         data_keys: &BTreeMap<ScVal, (StorageType, bool)>,
376 |         signing_keys: &[ed25519_dalek::SigningKey],
377 |     ) {
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19561 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (2 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

3 reference(s) to ECDSA across 2 file(s), e.g. soroban-env-host/src/cost_runner/cost_types/recover_ecdsa_secp256k1_key.rs:7. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	soroban-env-host/src/cost_runner/cost_types/recover_ecdsa_secp256k1_key.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:05 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open soroban-env-host/src/cost_runner/cost_types/recover_ecdsa_secp256k1_key.rs:7 and inspect the
quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · soroban-env-host/src/cost_runner/cost_types/recover_ecdsa_secp256k1_key.rs · line 7 · SHA-256 92080872f39f646c… · File 21256191032a43cd…

```
6 | };
7 | use k256::ecdsa::{RecoveryId, Signature};
8 |
```

file_excerpt · soroban-env-host/src/cost_runner/cost_types/recover_ecdsa_secp256k1_key.rs · line 28 · SHA-256 f3357cb7a595c1e3… · File 21256191032a43cd…

```
27 |             host.recover_key_ecdsa_secp256k1_internal(&sample.hash, &sample.sig, sample.rid)
28 |                 .expect("recover ecdsa secp256k1 key"),
29 |         );
```

file_excerpt · soroban-env-host/src/cost_runner/experimental/ecdsa_secp256k1_verify.rs · line 6 · SHA-256 58340089301d1dbf… · File d5d2bdc6133fc64c…

```
5 | };
6 | use k256::ecdsa::{Signature, VerifyingKey};
7 | use std::hint::black_box;
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19558 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: role not determined (9 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

19 reference(s) to Ed25519 across 9 file(s), e.g. soroban-env-host/src/builtin_contracts/testutils.rs:7. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:Ed25519:general
Data source	soroban-env-host/src/builtin_contracts/testutils.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:05 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open soroban-env-host/src/builtin_contracts/testutils.rs:7 and inspect the quoted lines; the
evidence lists further files.
```

EVIDENCE

file_excerpt · soroban-env-host/src/builtin_contracts/testutils.rs · line 7 · SHA-256 da5b8c5b9f906fbd… · File 59edd2300fad5707…

```
  6 | use crate::{Host, LedgerInfo};
  7 | use ed25519_dalek::{Signer, SigningKey};
  8 | use rand::Rng;
```

file_excerpt · soroban-env-host/src/cost_runner/cost_types/compute_ed25519_pubkey.rs · line 3 · SHA-256 74b96e0e9b3aa514… · File c679365a88252996…

```
  2 |
  3 | use ed25519_dalek::VerifyingKey;
  4 |
```

file_excerpt · soroban-env-host/src/cost_runner/cost_types/mod.rs · line 26 · SHA-256 f42d84c9a863f062… · File fd0538987c9bb5d5…

```
 25 | pub use bn254::*;
 26 | pub use compute_ed25519_pubkey::*;
 27 | pub use compute_keccak256_hash::*;
```

file_excerpt · soroban-env-host/src/cost_runner/cost_types/mod.rs · line 42 · SHA-256 6b9c07086467aba6… · File fd0538987c9bb5d5…

```
 41 | pub use verify_ecdsa_secp256r1_sig::*;
 42 | pub use verify_ed25519_sig::*;
 43 | pub use visit_object::*;
```

file_excerpt · soroban-env-host/src/cost_runner/cost_types/verify_ed25519_sig.rs · line 7 · SHA-256 27fd12545ba8cdbe… · File da86df368d3cdc76…

```
  6 | };
  7 | use ed25519_dalek::{Signature, VerifyingKey};
  8 |
```

file_excerpt · soroban-env-host/src/cost_runner/experimental/mod.rs · line 13 · SHA-256 50e2bafcfdae7aca… · File cc07f061fc69b534…

```
 12 | pub use ecdsa_secp256r1_recover::*;
 13 | pub use ed25519_scalar_mut::*;
 14 | pub use read_xdr::*;
```

file_excerpt · soroban-env-host/src/crypto/mod.rs · line 39 · SHA-256 cc46d772ead6ea56… · File cdfde68a22e3fc0a…

```
 38 |     // Ed25519 functions
 39 |     pub(crate) fn ed25519_signature_from_bytesobj_input(
 40 |         &self,
```

file_excerpt · soroban-env-host/src/crypto/mod.rs · line 43 · SHA-256 4bd187849dd9995a… · File cdfde68a22e3fc0a…

```
   42 |         sig: BytesObject,
   43 |     ) -> Result<ed25519_dalek::Signature, HostError> {
   44 |         self.fixed_length_bytes_from_bytesobj_input::<ed25519_dalek::Signature,
{ed25519_dalek::SIGNATURE_LENGTH}>(name, sig)
```

file_excerpt · soroban-env-host/src/crypto/mod.rs · line 44 · SHA-256 9d58b2c021c122e5… · File cdfde68a22e3fc0a…

```
   43 |     ) -> Result<ed25519_dalek::Signature, HostError> {
   44 |         self.fixed_length_bytes_from_bytesobj_input::<ed25519_dalek::Signature,
{ed25519_dalek::SIGNATURE_LENGTH}>(name, sig)
   45 |     }
```

file_excerpt · soroban-env-host/src/events/mod.rs · line 30 · SHA-256 3482bc89aebfeb8a… · File 8e8c489af9b03c97…

```
   29 |             PublicKeyTypeEd25519(e) => {
   30 |                 let strkey = stellar_strkey::ed25519::PublicKey(e.0);
   31 |                 write!(f, "{}", strkey)
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19559 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable BLS12-381: role not determined (5 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

9 reference(s) to BLS12-381 across 5 file(s), e.g. soroban-env-host/src/cost_runner/cost_types/bls12_381.rs:6. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:BLS12-381:general
Data source	soroban-env-host/src/cost_runner/cost_types/bls12_381.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:05 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open soroban-env-host/src/cost_runner/cost_types/bls12_381.rs:6 and inspect the quoted lines; the
evidence lists further files.
```

EVIDENCE

file_excerpt · soroban-env-host/src/cost_runner/cost_types/bls12_381.rs · line 6 · SHA-256 cbe62e9967b82476… · File 965dff98b1d17f2c…

```
    5 |     cost_runner::{CostRunner, CostType},
    6 |     crypto::bls12_381::FP_SERIALIZED_SIZE,
    7 |     impl_const_cost_runner_for_bls_consume_sample,
impl_const_cost_runner_for_bls_deref_sample,
```

file_excerpt · soroban-env-host/src/cost_runner/cost_types/mod.rs · line 1 · SHA-256 018bdeb4d2d82bbb… · File fd0538987c9bb5d5…

```
    1 | mod bls12_381;
    2 | mod bn254;
```

file_excerpt · soroban-env-host/src/cost_runner/cost_types/mod.rs · line 24 · SHA-256 1173daa967547ef6… · File fd0538987c9bb5d5…

```
   23 |
   24 | pub use bls12_381::*;
   25 | pub use bn254::*;
```

file_excerpt · soroban-env-host/src/cost_runner/experimental/mod.rs · line 1 · SHA-256 61704a40c8060cf4… · File cc07f061fc69b534…

```
    1 | mod bls12_381;
    2 | mod decode_secp256r1_sig;
```

file_excerpt · soroban-env-host/src/cost_runner/experimental/mod.rs · line 9 · SHA-256 38ff3b522b1267ca… · File cc07f061fc69b534…

```
    8 |
    9 | pub use bls12_381::*;
   10 | pub use decode_secp256r1_sig::*;
```

file_excerpt · soroban-env-host/src/crypto/mod.rs · line 23 · SHA-256 7990a12f3e16bfef… · File cdfde68a22e3fc0a…

```
   22 | use generic_array::ArrayLength;
   23 | pub(crate) mod bls12_381;
   24 | pub(crate) mod bn254;
```

file_excerpt · soroban-env-host/src/host.rs · line 3415 · SHA-256 18ef08fa0a3b37cc… · File 29dab11703256fe8…

```
3414 |
3415 |         if self.symbol_matches("BLS12_381".as_bytes(), field)? {
3416 |             self.poseidon_permutation_impl::<BlsScalar>(
```

file_excerpt · soroban-env-host/src/host.rs · line 3439 · SHA-256 89a8366163fe76c9… · File 29dab11703256fe8…

```
3438 |                 ScErrorCode::InvalidInput,
3439 |                 "poseidon_permutation: invalid field symbol, must be 'BLS12_381' or
```

```
'BN254'",
 3440 |                 &[field.to_val()],
```

file_excerpt · soroban-env-host/src/host.rs · line 3465 · SHA-256 6cc20e9377d6f354... · File 29dab11703256fe8...

```
 3464 |
 3465 |         if self.symbol_matches("BLS12_381".as_bytes(), field)? {
 3466 |             self.poseidon2_permutation_impl::<BlsScalar>(
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19560 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable BN254 (alt_bn128): role not determined (2 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

7 reference(s) to BN254 (alt_bn128) across 2 file(s), e.g. soroban-env-host/src/cost_runner/cost_types/bn254.rs:1. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:BN254 (alt_bn128):general
Data source	soroban-env-host/src/cost_runner/cost_types/bn254.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:05 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open soroban-env-host/src/cost_runner/cost_types/bn254.rs:1 and inspect the quoted lines; the
evidence lists further files.
```

EVIDENCE

file_excerpt · soroban-env-host/src/cost_runner/cost_types/bn254.rs · line 1 · SHA-256 be13ce1be21acf80… · File b77f33938ce6e3aa…

```
  1 | use ark_bn254::{Bn254, Fq, Fr, G1Affine, G1Projective, G2Affine};
  2 | use ark_ec::pairing::PairingOutput;
```

file_excerpt · soroban-env-host/src/cost_runner/cost_types/bn254.rs · line 120 · SHA-256 9a0f6e8dbd18df5b… · File b77f33938ce6e3aa…

```
119 |
120 | type InternalPairingOutput = PairingOutput<Bn254>;
121 | impl_lin_cost_runner_for_bls_deref_sample!(
```

file_excerpt · soroban-env-host/src/crypto/bn254.rs · line 4 · SHA-256 f7bf8b218395ce97… · File 91d5f195e217783d…

```
  3 |
  4 | use ark_bn254::{Bn254, Fq as Fp, Fq12, Fq2 as Fp2, Fr, G1Affine, G1Projective, G2Affine};
  5 | use ark_ec::AffineRepr;
```

file_excerpt · soroban-env-host/src/crypto/bn254.rs · line 526 · SHA-256 a4274c7c0a1836e3… · File 91d5f195e217783d…

```
525 |         vp2: &Vec<G2Affine>,
526 |     ) -> Result<PairingOutput<Bn254>, HostError> {
527 |         self.charge_budget(ContractCostType::Bn254Pairing, Some(vp1.len() as u64))?;
```

file_excerpt · soroban-env-host/src/crypto/bn254.rs · line 560 · SHA-256 05f2a34a6c508f13… · File 91d5f195e217783d…

```
559 |         // every panic condition has been excluded.
560 |         let mlo = Bn254::multi_miller_loop(vp1, vp2);
561 |         // final_exponentiation returning None means the `mlo.0.is_zero()`
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19562 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: role not determined (6 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

23 reference(s) to ECDSA across 6 file(s), e.g. soroban-env-host/src/cost_runner/cost_types/sec1_decode_point_uncompressed.rs:5. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDSA:general
Data source	soroban-env-host/src/cost_runner/cost_types/sec1_decode_point_uncompressed.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:05 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open soroban-env-host/src/cost_runner/cost_types/sec1_decode_point_uncompressed.rs:5 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · soroban-env-host/src/cost_runner/cost_types/sec1_decode_point_uncompressed.rs · line 5 · SHA-256 7d2cb0b1b110ea53… · File 06fda4cc5f09ca51…

```
4 | };
5 | use p256::ecdsa::VerifyingKey;
6 | use std::hint::black_box;
```

file_excerpt · soroban-env-host/src/cost_runner/cost_types/verify_ecdsa_secp256r1_sig.rs · line 5 · SHA-256 f2d53d01d18c1fcc… · File 8b1e249924c8f329…

```
4 | };
5 | use p256::ecdsa::{Signature, VerifyingKey};
6 | use std::hint::black_box;
```

file_excerpt · soroban-env-host/src/cost_runner/experimental/ecdsa_secp256r1_recover.rs · line 7 · SHA-256 f935f9980262fe70… · File 2e4920ce9df18301…

```
6 | use ecdsa::RecoveryId;
7 | use p256::ecdsa::Signature;
8 | use std::hint::black_box;
```

file_excerpt · soroban-env-host/src/cost_runner/experimental/sec1_decode_point_compressed.rs · line 5 · SHA-256 198adb0a3b79fd9b… · File 299089e64c18ca9c…

```
4 | };
5 | use p256::ecdsa::VerifyingKey;
6 |
```

file_excerpt · soroban-env-host/src/cost_runner/util.rs · line 12 · SHA-256 ae5512776b1eff8b… · File 9cbbfe0c50bd607d…

```
11 |         &self,
12 |         verifying_key: &k256::ecdsa::VerifyingKey,
13 |         msg_hash: &Hash,
```

file_excerpt · soroban-env-host/src/cost_runner/util.rs · line 22 · SHA-256 993641d576de59cd... · File 9cbbfe0c50bd607d...

```
21 |                     ScErrorCode::InvalidInput,
22 |                     "failed secp256r1 verification",
23 |                     &[],
```

file_excerpt · soroban-env-host/src/cost_runner/util.rs · line 35 · SHA-256 50fcd60c33b696e7... · File 9cbbfe0c50bd607d...

```
34 |         let recovered_key =
35 |             p256::ecdsa::VerifyingKey::recover_from_prehash(msg_hash.as_slice(), &sig, rid)
36 |                 .map_err(|_| {
```

file_excerpt · soroban-env-host/src/crypto/mod.rs · line 102 · SHA-256 d93603853957e179... · File cdfde68a22e3fc0a...

```
101 |         &self,
102 |         verifying_key: &p256::ecdsa::VerifyingKey,
103 |         msg_hash: &Hash,
```

file_excerpt · soroban-env-host/src/crypto/mod.rs · line 114 · SHA-256 24c43897781920ba... · File cdfde68a22e3fc0a...

```
113 |                     ScErrorCode::InvalidInput,
114 |                     "failed secp256r1 verification",
115 |                     &[],
```

file_excerpt · soroban-env-host/src/crypto/mod.rs · line 123 · SHA-256 fadb54cf6ddd4435... · File cdfde68a22e3fc0a...

```
122 |         bytes: &[u8],
123 |     ) -> Result<p256::ecdsa::VerifyingKey, HostError> {
124 |         use sec1::point::Tag;
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19563 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: authentication and identity (certificates, tokens, device attestation) (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

1 reference(s) to Ed25519 across 1 file(s), e.g. soroban-test-wasms/wasm-workspace/simple_account/src/lib.rs:33. Role, judged from paths, function names and nearby code: authentication and identity (certificates, tokens, device attestation). Short-lived authentication stays safe until a CRQC exists; long-lived identities (device attestation, CA and SSH keys) need a plan earlier.

Affected asset	source-code:Ed25519:authentication
Data source	soroban-test-wasms/wasm-workspace/simple_account/src/lib.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:05 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open soroban-test-wasms/wasm-workspace/simple_account/src/lib.rs:33 and inspect the quoted lines;
the evidence lists further files.
```

EVIDENCE

file_excerpt · soroban-test-wasms/wasm-workspace/simple_account/src/lib.rs · line 33 · SHA-256 304dbfe737276341… · File 7c74cf48ea7db5ed…

```
32 |         env.crypto()
33 |             .ed25519_verify(&public_key, &signature_payload.into(), &signature);
34 |     }
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19569 · DEP-KNOWN-ADVISORY

Medium

serde_with 3.12.0 is affected by GHSA-7gcf-g7xr-8hxj

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-7gcf-g7xr-8hxj for crates.io package serde_with at version 3.12.0: serde_with: KeyValueMap serialization panics on empty sequence or map entries. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/serde_with@3.12.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	33.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-7gcf-g7xr-8hxj
```

EVIDENCE

tool_output · OSV.dev GHSA-7gcf-g7xr-8hxj · manifest · SHA-256 9a6efbdf7670d9ea...


```
{"id": "GHSA-7gcf-g7xr-8hxj", "aliases": [], "modified": "2026-09-10T03:51:10.897685062Z",
"summary": "serde_with: KeyValueMap serialization panics on empty sequence or map entries",
"severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:L/SC:N/
SI:N/SA:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.21.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-7gcf-g7xr-8hxj

QSI-2026-19574 · DEP-KNOWN-ADVISORY

Medium

serde_with 3.14.0 is affected by GHSA-7gcf-g7xr-8hxj

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-7gcf-g7xr-8hxj for crates.io package serde_with at version 3.14.0: serde_with: KeyValueMap serialization panics on empty sequence or map entries. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/serde_with@3.14.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	33.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-7gcf-g7xr-8hxj
```

EVIDENCE

tool_output · OSV.dev GHSA-7gcf-g7xr-8hxj · manifest · SHA-256 9a6efbdf7670d9ea...


```
{"id": "GHSA-7gcf-g7xr-8hxj", "aliases": [], "modified": "2026-09-10T03:51:10.897685062Z",
"summary": "serde_with: KeyValueMap serialization panics on empty sequence or map entries",
"severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:L/SC:N/
SI:N/SA:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.21.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-7gcf-g7xr-8hxj

QSI-2026-19570 · DEP-KNOWN-ADVISORY

Medium

time 0.3.41 is affected by GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009) for crates.io package time at version 0.3.41: time vulnerable to stack exhaustion Denial of Service attack. The audited code does not name the affected functions (time::Date::parse, time::OffsetDateTime::parse, time::PrimitiveDateTime::parse, time::Time::parse, time::UtcDateTime::parse, time::UtcOffset::parse ...); time comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/time@0.3.41
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	28.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-r6v5-fh4h-64xc
```

EVIDENCE

tool_output · OSV.dev GHSA-r6v5-fh4h-64xc · manifest · SHA-256 47bed44226c6f8f5…


```
{"id": "GHSA-r6v5-fh4h-64xc", "aliases": ["CVE-2026-25727", "RUSTSEC-2026-0009"], "modified":
"2026-09-10T03:50:35.583241382Z", "summary": "time vulnerable to stack exhaustion Denial of Service
attack", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:H/AT:N/PR:L/UI:A/VC:N/VI:N/
VA:H/SC:N/SI:N/SA:H"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.3.47).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-r6v5-fh4h-64xc

QSI-2026-19568 · DEP-KNOWN-ADVISORY

Low

rand 0.8.5 is affected by GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097) for crates.io package rand at version 0.8.5: Rand is unsound with a custom logger using rand::rng(). The audited code names rand::rng: the affected code may be reachable.

Affected asset	dependency:crates.io/rand@0.8.5
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	39.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-cq8v-f236-94qc
```

EVIDENCE

tool_output · OSV.dev GHSA-cq8v-f236-94qc · manifest · SHA-256 a61cde8ef254431f...


```
{"id": "GHSA-cq8v-f236-94qc", "aliases": ["RUSTSEC-2026-0097"], "modified":
"2026-09-10T03:50:43.958966498Z", "summary": "Rand is unsound with a custom logger using
rand::rng()", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.10.1, 0.8.6, 0.9.3).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-cq8v-f236-94qc

QSI-2026-19571 · DEP-KNOWN-ADVISORY

Low

tracing-subscriber 0.3.18 is affected by GHSA-xwfj-jgwm-7wp5 (CVE-2025-58160, RUSTSEC-2025-0055)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-xwfj-jgwm-7wp5 (CVE-2025-58160, RUSTSEC-2025-0055) for crates.io package tracing-subscriber at version 0.3.18: Tracing logging user input may result in poisoning logs with ANSI escape sequences. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/tracing-subscriber@0.3.18
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	29.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-xwfj-jgwm-7wp5
```

EVIDENCE

tool_output · OSV.dev GHSA-xwfj-jgwm-7wp5 · manifest · SHA-256 c5ea2040929b0bf0...


```
{"id": "GHSA-xwfj-jgwm-7wp5", "aliases": ["CVE-2025-58160", "RUSTSEC-2025-0055"], "modified":
"2026-09-10T03:50:27.766863705Z", "summary": " Tracing logging user input may result in poisoning
logs with ANSI escape sequences", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:L/
AT:P/PR:N/UI:P/VC:N/VI:L/VA:N/SC:N/SI:N/SA:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.3.20).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-xwfj-jgwm-7wp5

QSI-2026-19566 · DEP-KNOWN-ADVISORY

Low

keccak 0.1.4 is affected by GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012) for crates.io package keccak at version 0.1.4: Unsoundness in opt-in ARMv8 assembly backend for `keccak`. The affected code sits behind the `asm` Cargo feature, off by default, which no audited manifest enables: the advisory does not apply unless another dependency enables it.

Affected asset	dependency:crates.io/keccak@0.1.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	25.0 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-3288-p39f-rqpv
```

EVIDENCE

tool_output · OSV.dev GHSA-3288-p39f-rqpv · manifest · SHA-256 d5a8f5565101ab9d...


```
{"id": "GHSA-3288-p39f-rqpv", "aliases": ["RUSTSEC-2026-0012"], "modified":
"2026-09-10T03:50:59.332505686Z", "summary": "Unsoundness in opt-in ARMv8 assembly backend for
```

`keccak`", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:H/AT:P/PR:N/UI:N/VC:N/VI:L/VA:N/SC:N/SI:N/SA:N/E:U"}], "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.1.6).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-3288-p39f-rqpv

QSI-2026-19564 · DEP-ADVISORY-UNRATED

Low

crypto-bigint 0.5.2 is affected by RUSTSEC-2026-0387 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0387 for crates.io package crypto-bigint at version 0.5.2: Multiple timing variability issues in operations intended to be constant-time. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/crypto-bigint@0.5.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0387
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0387 · manifest · SHA-256 eab63427da988e95…

{"id": "RUSTSEC-2026-0387", "aliases": [], "modified": "2026-10-24T03:45:02.365137418Z", "summary": "Multiple timing variability issues in operations intended to be constant-time", "severity": null, "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo

feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.7.6) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0387

QSI-2026-19565 · DEP-ADVISORY-UNRATED

Low

ctutils 0.4.2 is affected by RUSTSEC-2026-0385 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0385 for crates.io package ctutils at version 0.4.2: Non-constant-time mask generation and `u128` comparisons. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/ctutils@0.4.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0385
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0385 · manifest · SHA-256 200470326c8a5ad1...


```
{"id": "RUSTSEC-2026-0385", "aliases": [], "modified": "2026-10-23T08:30:02.389575143Z", "summary":
"Non-constant-time mask generation and `u128` comparisons", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.4.4) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0385

QSI-2026-19572 · DEP-ADVISORY-UNRATED

Low

crypto-bigint 0.5.4 is affected by RUSTSEC-2026-0387 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0387 for crates.io package crypto-bigint at version 0.5.4: Multiple timing variability issues in operations intended to be constant-time. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/crypto-bigint@0.5.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0387
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0387 · manifest · SHA-256 eab63427da988e95...


```
{"id": "RUSTSEC-2026-0387", "aliases": [], "modified": "2026-10-24T03:45:02.365137418Z", "summary":
"Multiple timing variability issues in operations intended to be constant-time", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.7.6) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0387

QSI-2026-19567 · DEP-ADVISORY-INFORMATIONAL

Info

paste 1.0.12: informational advisory RUSTSEC-2024-0436 (unmaintained)

Improvement recommendation | Confidence: Low

OSV.dev lists RUSTSEC-2024-0436 for crates.io package paste at version 1.0.12 as informational (unmaintained): paste - no longer maintained. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/paste@1.0.12
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	11.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2024-0436
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2024-0436 · manifest · SHA-256 ea867adb8ba26c96...


```
{"id": "RUSTSEC-2024-0436", "aliases": [], "modified": "2026-01-27T22:03:15.694766Z", "summary":
"paste - no longer maintained", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2024-0436

QSI-2026-19573 · DEP-ADVISORY-INFORMATIONAL

Info

paste 1.0.14: informational advisory RUSTSEC-2024-0436 (unmaintained)

Improvement recommendation | Confidence: Low

OSV.dev lists RUSTSEC-2024-0436 for crates.io package paste at version 1.0.14 as informational (unmaintained): paste - no longer maintained. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/paste@1.0.14
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.

Analyzer	dependencies 1.3.0 · 2026-11-06 16:05 UTC
Priority score	11.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2024-0436
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2024-0436 · manifest · SHA-256 ea867adb8ba26c96...


```
{"id": "RUSTSEC-2024-0436", "aliases": [], "modified": "2026-01-27T22:03:15.694766Z", "summary":
"paste - no longer maintained", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2024-0436

Evidence

ID	Type	Data source	SHA-256
QSI-2026-19557	file_excerpt	soroban-env-host/src/builtin_contracts/stellar_asset_contract/metadata.rs (line 158)	3d7fb8af17a8920210188c6a4caffc601f7bc324ca0d4bb4d
QSI-2026-19557	file_excerpt	soroban-env-host/src/testutils.rs (line 15)	c8d71454e9e88769c00e358ff438c6a9feb5ef52d447c80c4
QSI-2026-19557	file_excerpt	soroban-env-host/src/testutils.rs (line 316)	ccca624ff7d49ed38d518548cc6e466cad617e88f73ada212
QSI-2026-19557	file_excerpt	soroban-env-host/src/testutils.rs (line 376)	5c948ca4d4acf0e68fb3ff6bfee3dd06b880aa4012d1dd8ac
QSI-2026-19561	file_excerpt	soroban-env-host/src/cost_runner/cost_types/recover_ecdsa_secp256k1_key.rs (line 7)	92080872f39f646ca72af7f62a1a86aafc23f66369b0c021b
QSI-2026-19561	file_excerpt	soroban-env-host/src/cost_runner/cost_types/recover_ecdsa_secp256k1_key.rs (line 28)	f3357cb7a595c1e3ce6deec4fed0973bae3735ea3092adff5
QSI-2026-19561	file_excerpt	soroban-env-host/src/cost_runner/experimental/ecdsa_secp256k1_verify.rs (line 6)	58340089301d1dbfa155c975109ed0bf7958615b3c0a53cda
QSI-2026-19558	file_excerpt	soroban-env-host/src/builtin_contracts/testutils.rs (line 7)	da5b8c5b9f906fbda229feb92e848a55c67100ba443c96cd7
QSI-2026-19558	file_excerpt	soroban-env-host/src/cost_runner/cost_types/compute_ed25519_pubkey.rs (line 3)	74b96e0e9b3aa5141b6dd1b7ed5f7590736faade905cb0fa9
QSI-2026-19558	file_excerpt	soroban-env-host/src/cost_runner/cost_types/mod.rs (line 26)	f42d84c9a863f0627967b5f03b249c98b9e410073e1e88532
QSI-2026-19558	file_excerpt	soroban-env-host/src/cost_runner/cost_types/mod.rs (line 42)	6b9c07086467aba6a28db1678918325cc31c972f1e0c79307
QSI-2026-19558	file_excerpt	soroban-env-host/src/cost_runner/cost_types/verify_ed25519_sig.rs (line 7)	27fd12545ba8cdbed39e35e8e33d4ec4344551df83db28052
QSI-2026-19558	file_excerpt	soroban-env-host/src/cost_runner/experimental/mod.rs (line 13)	50e2bafcfdae7acadbc0e94bfa00f85ca5b7f4a2639b0518c
QSI-2026-19558	file_excerpt	soroban-env-host/src/crypto/mod.rs (line 39)	cc46d772ead6ea569a6eb081e8122217b5cd15b0e9ff55468
QSI-2026-19558	file_excerpt	soroban-env-host/src/crypto/mod.rs (line 43)	4bd187849dd9995abe44a09658e6908138d52b8fc02de9fcf

QSI-2026-19558	file_excerpt	soroban-env-host/src/crypto/mod.rs (line 44)	9d58b2c021c122e57885affb53f02fd2a5fc0cd59817ba5c4
QSI-2026-19558	file_excerpt	soroban-env-host/src/events/mod.rs (line 30)	3482bc89aebfeb8ad0d9b9651d72695d317c56f48b7ac7efa
QSI-2026-19559	file_excerpt	soroban-env-host/src/cost_runner/cost_types/bls12_381.rs (line 6)	cbe62e9967b82476495c9289979771bf8833f7794b9da7dff
QSI-2026-19559	file_excerpt	soroban-env-host/src/cost_runner/cost_types/mod.rs (line 1)	018bdeb4d2d82bbb73383000c031377b65daa884f6aa06076
QSI-2026-19559	file_excerpt	soroban-env-host/src/cost_runner/cost_types/mod.rs (line 24)	1173daa967547ef6f92015858df988ca204347e3b0c86165f
QSI-2026-19559	file_excerpt	soroban-env-host/src/cost_runner/experimental/mod.rs (line 1)	61704a40c8060cf429540fade251e2c1e888ad9950384d994
QSI-2026-19559	file_excerpt	soroban-env-host/src/cost_runner/experimental/mod.rs (line 9)	38ff3b522b1267ca4d3296c833e5512d6bc78f147cee0494e
QSI-2026-19559	file_excerpt	soroban-env-host/src/crypto/mod.rs (line 23)	7990a12f3e16bfef9c4846c4ef99db1fac6a5217e5e22db8f
QSI-2026-19559	file_excerpt	soroban-env-host/src/host.rs (line 3415)	18ef08fa0a3b37cc3441d8d4bb1466aa33065b16ac19f8229
QSI-2026-19559	file_excerpt	soroban-env-host/src/host.rs (line 3439)	89a8366163fe76c93b72608f1660664a4d76ad4083e851695
QSI-2026-19559	file_excerpt	soroban-env-host/src/host.rs (line 3465)	6cc20e9377d6f35423e985acb0fa303ed0a23c3d796e183e7
QSI-2026-19560	file_excerpt	soroban-env-host/src/cost_runner/cost_types/bn254.rs (line 1)	be13ce1be21acf8085c5bf580cbd12293aa118ec05a6bb03c
QSI-2026-19560	file_excerpt	soroban-env-host/src/cost_runner/cost_types/bn254.rs (line 120)	9a0f6e8dbd18df5bc4f14ddd7dde622d23472fa85a9cb2a20
QSI-2026-19560	file_excerpt	soroban-env-host/src/crypto/bn254.rs (line 4)	f7bf8b218395ce977ab90e65b7e49029f29344afa5b619dac
QSI-2026-19560	file_excerpt	soroban-env-host/src/crypto/bn254.rs (line 526)	a4274c7c0a1836e3effc17a944587fc4523bc97379c56241b
QSI-2026-19560	file_excerpt	soroban-env-host/src/crypto/bn254.rs (line 560)	05f2a34a6c508f13012111c63e0e6e88430ec4b4a066bf471
QSI-2026-19562	file_excerpt	soroban-env-host/src/cost_runner/cost_types/sec1_decode_point_uncompressed.rs (line 5)	7d2cb0b1b110ea53e52a6a1c0f0e5651f2b202b8ee2d8a87a
QSI-2026-19562	file_excerpt	soroban-env-host/src/cost_runner/cost_types/verify_ecdsa_secp256r1_sig.rs (line 5)	f2d53d01d18c1fcc2292c319846326c466197df63a11674ca
QSI-2026-19562	file_excerpt	soroban-env-host/src/cost_runner/experimental/ecdsa_secp256r1_recover.rs (line 7)	f935f9980262fe702e4e637a6cf0d94ad4f11c566858c9858

QSI-2026-19562	file_excerpt	soroban-env-host/src/cost_runner/experimental/sec1_decode_point_compressed.rs (line 5)	198adb0a3b79fd9b102dd1d8010f7ec744f8079e2a92651a7
QSI-2026-19562	file_excerpt	soroban-env-host/src/cost_runner/util.rs (line 12)	ae5512776b1eff8b7ccfcfb61c0d70042d1f24134c5c6eda6
QSI-2026-19562	file_excerpt	soroban-env-host/src/cost_runner/util.rs (line 22)	993641d576de59cd5cd5c46079bf94b0b97dd9b3daf2fa32a
QSI-2026-19562	file_excerpt	soroban-env-host/src/cost_runner/util.rs (line 35)	50fcd60c33b696e792d7360322abca7b4a8d38f256054a059
QSI-2026-19562	file_excerpt	soroban-env-host/src/crypto/mod.rs (line 102)	d93603853957e179e153b1c85f5515848dab1bb4afe578505
QSI-2026-19562	file_excerpt	soroban-env-host/src/crypto/mod.rs (line 114)	24c43897781920baa5c997781ba7250fbf2159ab956683235
QSI-2026-19562	file_excerpt	soroban-env-host/src/crypto/mod.rs (line 123)	fadb54cf6ddd44355e4ad2b67f56ed06be857d0c58fb0c9a5
QSI-2026-19563	file_excerpt	soroban-test-wasms/wasm-workspace/simple_account/src/lib.rs (line 33)	304dbfe7372763418ad046fb11f27c77b2f1ff85af31a3044
QSI-2026-19569	tool_output	OSV.dev GHSA-7gcf-g7xr-8hxj (manifest)	9a6efbdf7670d9ead9898fb39da9d141ceec2380712ba7d56
QSI-2026-19574	tool_output	OSV.dev GHSA-7gcf-g7xr-8hxj (manifest)	9a6efbdf7670d9ead9898fb39da9d141ceec2380712ba7d56
QSI-2026-19570	tool_output	OSV.dev GHSA-r6v5-fh4h-64xc (manifest)	47bed44226c6f8f5c94bfd27c3dd67ade62aa708c5d61bcca
QSI-2026-19568	tool_output	OSV.dev GHSA-cq8v-f236-94qc (manifest)	a61cde8ef254431f750cff43ce8ebe999ba61525990761c3c
QSI-2026-19571	tool_output	OSV.dev GHSA-xwfj-jgwm-7wp5 (manifest)	c5ea2040929b0bf0d590c444d6ef1ef2de439865d4d25d158
QSI-2026-19566	tool_output	OSV.dev GHSA-3288-p39f-rqpv (manifest)	d5a8f5565101ab9d60521142f31c54583db5b87e9a59fc99e
QSI-2026-19564	tool_output	OSV.dev RUSTSEC-2026-0387 (manifest)	eab63427da988e95becf061fd8169abcb9634f743aeeb7567
QSI-2026-19565	tool_output	OSV.dev RUSTSEC-2026-0385 (manifest)	200470326c8a5ad19deb3639dff41d33749b1153097b6b87e
QSI-2026-19572	tool_output	OSV.dev RUSTSEC-2026-0387 (manifest)	eab63427da988e95becf061fd8169abcb9634f743aeeb7567
QSI-2026-19567	tool_output	OSV.dev RUSTSEC-2024-0436 (manifest)	ea867adb8ba26c96c3779becf55b3de25d46f37f23ab55131
QSI-2026-19573	tool_output	OSV.dev RUSTSEC-2024-0436 (manifest)	ea867adb8ba26c96c3779becf55b3de25d46f37f23ab55131

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 2 of 4 — post-quantum in use outside the critical paths

- In use: ML-DSA — soroban-env-host/src/cost_runner/cost_types/mod.rs
- In use: ML-DSA — soroban-env-host/src/host.rs
- Present, not wired: ML-DSA — soroban-env-host/src/cost_runner/cost_types/ml_dsa.rs
- Present, not wired: ML-DSA — soroban-env-host/src/crypto/ml_dsa.rs
- A post-quantum library is declared as a dependency

Next step: Extend post-quantum protection to the critical roles that stay classical, starting with what can be recorded today (wrapped keys, channel key exchange).

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
transaction and message signing	ECDSA (2), Ed25519 (2)	QSI-2026-19557, QSI-2026-19561	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them
authentication and identity (certificates, tokens, device attestation)	Ed25519 (1)	QSI-2026-19563	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them

role not determined	Ed25519 (9), ECDSA (6), BLS12-381 (5), BN254 (alt_bn128) (2)	QSI-2026-19558, QSI-2026-19559, QSI-2026-19560, QSI-2026-19562	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
reduced margin (Grover), approved	1
Grover: acceptable to NIST, below ECCG/ANSSI guidance	15
n/a	16
post-quantum	6
vulnerable (Shor)	38

Detailed findings: QSI-2026-19557, QSI-2026-19561, QSI-2026-19558, QSI-2026-19559, QSI-2026-19560, QSI-2026-19562, QSI-2026-19563

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — discrete logarithms in finite fields Q1: quantum computation offline, classical access to the victim	breaks	BLS12-381 (7), BN254 (alt_bn128) (2)	Breaks finite-field Diffie–Hellman and DSA at every size. Move key establishment to ML-KEM (hybrid with X25519 or ECDH during the transition) and signatures to ML-DSA or SLH-DSA.
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	BLS12-381 (7), BN254 (alt_bn128) (2), ECDSA (16), Ed25519 (13)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.

Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	HMAC (1), Keccak-256 (2), Poseidon (8), SHA-2 (5)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	Keccak-256 (2), Poseidon (8), SHA-2 (5)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.
Quantum lattice sieving Q1: quantum computation offline, classical access to the victim	included in standard parameters	ML-DSA (4)	No change for ML-KEM or ML-DSA at the standard parameter sets. A 2024 claim of a polynomial-time quantum LWE algorithm was withdrawn after a bug was found (see claims).

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.
- osv_advisories — not checked: 3 package(s) are built from git repositories (forks or unreleased code), so registry advisories were not looked up for them: soroban-wasmi, wasmi_arena, wasmi_core

Remediation plan

QSI-2026-19569 · P2 · 30 days

root-cause fix

Upgrade serde_with to a fixed version (3.21.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-7gcf-g7xr-8hxj.

QSI-2026-19574 · P2 · 30 days

root-cause fix

Upgrade serde_with to a fixed version (3.21.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-7gcf-g7xr-8hxj.

QSI-2026-19570 · P2 · 30 days

root-cause fix

Upgrade time to a fixed version (0.3.47).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009).

QSI-2026-19568 · P2 · 30 days

root-cause fix

Upgrade rand to a fixed version (0.10.1, 0.8.6, 0.9.3).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low

Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097).

QSI-2026-19571 · P2 · 30 days

root-cause fix

Upgrade tracing-subscriber to a fixed version (0.3.20).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-xwfj-jgwm-7wp5 (CVE-2025-58160, RUSTSEC-2025-0055).

QSI-2026-19566 · P2 · 30 days

root-cause fix

Upgrade keccak to a fixed version (0.1.6).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012).

QSI-2026-19564 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	

	Dependency scan no longer reports RUSTSEC-2026-0387, or the triage is recorded.

QSI-2026-19565 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0385; upgrade ctutils to a fixed version (0.4.4) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0385, or the triage is recorded.

QSI-2026-19572 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0387, or the triage is recorded.

QSI-2026-19557 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19561 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19558 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19559 · P3 · 90 days

short-term risk reduction

Add BLS12-381 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19560 · P3 · 90 days

short-term risk reduction

Add BN254 (alt_bn128) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19562 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19563 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19557 · P4 · 90 days root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19561 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19558 · P4 · 90 days root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	

	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19559 · P4 · 90 days

root-cause fix

Plan migration of BLS12-381 to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-19560 · P4 · 90 days

root-cause fix

Plan migration of BN254 (alt_bn128) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-19562 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19563 · P4 · 90 days root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19567 · P4 · 90 days root-cause fix

Replace paste with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	paste replaced, or the decision recorded.

QSI-2026-19573 · P4 · 90 days root-cause fix

Replace paste with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	paste replaced, or the decision recorded.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
P2	QSI-2026-19569	Upgrade serde_with to a fixed version (3.21.0).	root-cause fix
P2	QSI-2026-19574	Upgrade serde_with to a fixed version (3.21.0).	root-cause fix
P2	QSI-2026-19570	Upgrade time to a fixed version (0.3.47).	root-cause fix
P2	QSI-2026-19568	Upgrade rand to a fixed version (0.10.1, 0.8.6, 0.9.3).	root-cause fix
P2	QSI-2026-19571	Upgrade tracing-subscriber to a fixed version (0.3.20).	root-cause fix
P2	QSI-2026-19566	Upgrade keccak to a fixed version (0.1.6).	root-cause fix

60 days

Priority	ID	Action	Type
P3	QSI-2026-19564	Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.	root-cause fix
P3	QSI-2026-19565	Triage RUSTSEC-2026-0385; upgrade ctutils to a fixed version (0.4.4) if it applies.	root-cause fix
P3	QSI-2026-19572	Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.	root-cause fix

90 days

Priority	ID	Action	Type
P3	QSI-2026-19557	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19561	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19558	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19559	Add BLS12-381 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19560	Add BN254 (alt_bn128) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19562	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction

P3	QSI-2026-19563	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-19557	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19561	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19558	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19559	Plan migration of BLS12-381 to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19560	Plan migration of BN254 (alt_bn128) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19562	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19563	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19567	Replace paste with a maintained alternative or record why it stays.	root-cause fix
P4	QSI-2026-19573	Replace paste with a maintained alternative or record why it stays.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-19557	Re-run the inventory after refactoring.
QSI-2026-19561	Re-run the inventory after refactoring.
QSI-2026-19558	Re-run the inventory after refactoring.
QSI-2026-19559	Re-run the inventory after refactoring.
QSI-2026-19560	Re-run the inventory after refactoring.
QSI-2026-19562	Re-run the inventory after refactoring.
QSI-2026-19563	Re-run the inventory after refactoring.
QSI-2026-19569	Re-run the dependency scan after upgrading.
QSI-2026-19574	Re-run the dependency scan after upgrading.
QSI-2026-19570	Re-run the dependency scan after upgrading.
QSI-2026-19568	Re-run the dependency scan after upgrading.
QSI-2026-19571	Re-run the dependency scan after upgrading.
QSI-2026-19566	Re-run the dependency scan after upgrading.
QSI-2026-19564	Re-run the dependency scan after upgrading.
QSI-2026-19565	Re-run the dependency scan after upgrading.
QSI-2026-19572	Re-run the dependency scan after upgrading.
QSI-2026-19567	Re-run the dependency scan after the change.
QSI-2026-19573	Re-run the dependency scan after the change.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.2.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.3.0
inventory_document	1.1.0
solidity	1.3.0
source_code	1.4.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.3.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

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8. Common Vulnerability Scoring System v3.1: Specification Document — FIRST. https://www.first.org/cvss/v3.1/specification-document (final)
9. D. J. Bernstein — Cost analysis of hash collisions: Will quantum computers make SHARCS obsolete? (2009) — cr.yp.to. https://cr.yp.to/hash/collisioncost-20090823.pdf (paper dated 2009-08-23)
10. D. Litinski — How to compute a 256-bit elliptic curve private key with only 50 million Toffoli gates (arXiv:2306.08585) — arXiv. https://arxiv.org/abs/2306.08585 (preprint (2023-06-14): 6000 modules × 1152 physical qubits, one key per 10 minutes under its architecture assumptions)
11. FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM) — NIST. https://csrc.nist.gov/pubs/fips/203/final (final (August 2024))
12. FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA) — NIST. https://csrc.nist.gov/pubs/fips/204/final (final (August 2024))
13. FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA) — NIST. https://csrc.nist.gov/pubs/fips/205/final (final (August 2024))
14. G. Brassard, P. Høyer, A. Tapp — Quantum Algorithm for the Collision Problem (LATIN '98; arXiv:quant-ph/9705002) — Springer / arXiv. https://arxiv.org/abs/quant-ph/9705002 (peer-reviewed article: collisions of r-to-one functions in O((N/r)^{1/3}) expected evaluations)
15. J. Proos, C. Zalka — Shor's discrete logarithm quantum algorithm for elliptic curves (Quantum Inf. Comput. 3(4), 2003; arXiv:quant-ph/0301141) — Quantum Information & Computation. https://doi.org/10.26421/QIC3.4-3 (peer-reviewed article)
16. L. K. Grover — A fast quantum mechanical algorithm for database search (STOC 1996) — ACM. https://doi.org/10.1145/237814.237866 (peer-reviewed article)
17. M. Amy et al. — Estimating the cost of generic quantum pre-image attacks on SHA-2 and SHA-3 (SAC 2016; arXiv:1603.09383) — Springer / arXiv. https://arxiv.org/abs/1603.09383 (peer-reviewed article: SHA-256 about 2^153.8 surface-code cycles on 2^12.6 logical qubits; SHA3-256 about 2^146.5 cycles on 2^20)
18. M. Cain et al. — Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits (arXiv:2603.28627) — arXiv. https://arxiv.org/abs/2603.28627 (preprint (2026-03-30): P-256 in a few days with 26,000 physical neutral-atom qubits; RSA-2048 one to two orders of magnitude longer)
19. M. Ekerå, J. Gärtner — A high-level comparison of state-of-the-art quantum algorithms for breaking asymmetric cryptography (arXiv:2405.14381) — arXiv. https://arxiv.org/abs/2405.14381 (preprint (2024-05-23): Regev-style algorithms show no overall advantage yet for cryptographic RSA and finite-field sizes)
20. M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018) — IEEE. https://doi.org/10.1109/MSP.2018.3761723 (peer-reviewed article; source of the x + y > z migration inequality)

21. M. Roetteler, M. Naehrig, K. Svore, K. Lauter — Quantum resource estimates for computing elliptic curve discrete logarithms (ASIACRYPT 2017) — IACR / arXiv. https://arxiv.org/abs/1706.06752 (peer-reviewed article: at most 9n + 2⌈log2 n⌉ + 10 qubits and 448 n^3 log2 n + 4090 n^3 Toffoli gates)
22. NIST IR 8547 — Transition to Post-Quantum Cryptography Standards — NIST. https://csrc.nist.gov/pubs/ir/8547/final (final (2026-10-27; the initial public draft of November 2024 is obsoleted). Tables 2 and 4: ECDSA, EdDSA and RSA signatures, finite-field and elliptic-curve DH and MQV, and RSA key establishment are deprecated after 2030 and disallowed after 2035 at every key size — the draft deprecated only the 112-bit level. A composite (hybrid) key establishment stays FIPS 140-validatable while one input comes from an approved scheme. Repeats the federal deadlines of OMB M-26-15: post-quantum key establishment by 2030-12-31 and signatures by 2031-12-31 for high value assets and high impact systems)
23. NIST — Post-Quantum Cryptography Standardization: Evaluation Criteria, Security (Call for Proposals, Section 4.A.5) — NIST. https://csrc.nist.gov/Projects/Post-Quantum-Cryptography/Post-Quantum-Cryptography-Standardization/Evaluation-Criteria/Security-(Evaluation-Criteria) (security categories: AES-128 key search 2^170/MAXDEPTH quantum gates, AES-192 2^233/MAXDEPTH, AES-256 2^298/MAXDEPTH; MAXDEPTH 2^40–2^96)
24. OSV advisory GHSA-3288-p39f-rqpv — OSV.dev. https://osv.dev/vulnerability/GHSA-3288-p39f-rqpv (modified 2026-09-10T03:50:59.332505686Z)
25. OSV advisory GHSA-7gcf-g7xr-8hxj — OSV.dev. https://osv.dev/vulnerability/GHSA-7gcf-g7xr-8hxj (modified 2026-09-10T03:51:10.897685062Z)
26. OSV advisory GHSA-cq8v-f236-94qc — OSV.dev. https://osv.dev/vulnerability/GHSA-cq8v-f236-94qc (modified 2026-09-10T03:50:43.958966498Z)
27. OSV advisory GHSA-r6v5-fh4h-64xc — OSV.dev. https://osv.dev/vulnerability/GHSA-r6v5-fh4h-64xc (modified 2026-09-10T03:50:35.583241382Z)
28. OSV advisory GHSA-xwfj-jgwm-7wp5 — OSV.dev. https://osv.dev/vulnerability/GHSA-xwfj-jgwm-7wp5 (modified 2026-09-10T03:50:27.766863705Z)
29. OSV advisory RUSTSEC-2024-0436 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2024-0436 (modified 2026-01-27T22:03:15.694766Z)
30. OSV advisory RUSTSEC-2026-0385 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0385 (modified 2026-10-23T08:30:02.389575143Z)
31. OSV advisory RUSTSEC-2026-0387 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0387 (modified 2026-10-24T03:45:02.365137418Z)
32. P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997) — SIAM. https://doi.org/10.1137/S0097539795293172 (peer-reviewed article)
33. R. Babbush et al. — Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations (arXiv:2603.28846) — arXiv. https://arxiv.org/abs/2603.28846 (preprint (2026-03-30, v2 2026-04-15): 256-bit ECDLP with < 1200 logical qubits and < 90 M Toffoli gates (or < 1450 and < 70 M); < 500,000 physical qubits at 10^-3 error, minutes)
34. S. Jaques, M. Naehrig, M. Roetteler, F. Virdia — Implementing Grover oracles for quantum key search on AES and LowMC (EUROCRYPT 2020; arXiv:1910.01700) — Springer / arXiv. https://arxiv.org/abs/1910.01700 (peer-reviewed article: lower cost estimates for the AES key searches that define the NIST categories)

Appendices

A. Cryptographic inventory

Algorithm	Parameter	Quantum status	Classical status	Count	Locations	Examples
HMAC		reduced margin (Grover), approved	acceptable	1	1	soroban-env-host/src/crypto/mod.rs:14
Keccak-256	Keccak-256	Grover: acceptable to NIST, below ECCG/ ANSSI guidance	acceptable	2	2	soroban-env-host/src/cost_runner/ cost_types/ compute_keccak256_hash.rs; soroban-env-host/src/crypto/mod.rs:18
Poseidon		Grover: acceptable to NIST, below ECCG/ ANSSI guidance	acceptable	8	8	soroban-env-host/src/crypto/mod.rs:5; soroban-env-host/src/crypto/poseidon/ mod.rs:6; soroban-env-host/src/crypto/ poseidon/poseidon.rs:4
SHA-2	SHA-256	Grover: acceptable to NIST, below ECCG/ ANSSI guidance	acceptable	5	5	soroban-env-host/src/cost_runner/ cost_types/compute_sha256_hash.rs; soroban-env-host/src/crypto/ bls12_381.rs:27; soroban-env-host/src/ crypto/mod.rs:17
Crypto library p256 (RustCrypto elliptic curves)	0.13.2	n/a	acceptable	3	3	Cargo.lock; soroban-env-host/ Cargo.toml; soroban-test-wasms/wasm-workspace/Cargo.lock
Crypto library k256 (RustCrypto elliptic curves)	0.13.4	n/a	acceptable	2	2	Cargo.lock; soroban-test-wasms/wasm-workspace/Cargo.lock
Crypto library ecdsa (python-ecdsa)	0.16.9	n/a	acceptable	2	2	Cargo.lock; soroban-test-wasms/wasm-workspace/Cargo.lock

Crypto library ed25519-dalek (ed25519-dalek)	2.1.1	n/a	acceptable	1	1	Cargo.lock
Crypto library curve25519-dalek (ed25519-dalek)	4.1.3	n/a	acceptable	2	2	Cargo.lock; soroban-test-wasms/wasm-workspace/Cargo.lock
Crypto library k256 (RustCrypto elliptic curves)	0.13.1	n/a	acceptable	1	1	soroban-env-host/Cargo.toml
Crypto library k256 (RustCrypto elliptic curves)	0.13.3	n/a	acceptable	1	1	soroban-env-host/Cargo.toml
Crypto library ecdsa (python-ecdsa)	0.16.7	n/a	acceptable	1	1	soroban-env-host/Cargo.toml
Crypto library ed25519-dalek (ed25519-dalek)	2.0.0	n/a	acceptable	2	2	soroban-env-host/Cargo.toml; soroban-test-wasms/wasm-workspace/Cargo.lock
Crypto library curve25519-dalek (ed25519-dalek)	>=4.1.3	n/a	acceptable	1	1	soroban-env-host/Cargo.toml
ML-DSA		post-quantum	acceptable	4	4	soroban-env-host/src/cost_runner/cost_types/ml_dsa.rs:9; soroban-env-host/src/cost_runner/cost_types/mod.rs:34; soroban-env-host/src/crypto/ml_dsa.rs:21

PQC library ml-dsa (RustCrypto post-quantum crates)	0.1.1	post-quantum	acceptable	2	2	Cargo.lock; soroban-env-host/Cargo.toml
BLS12-381		vulnerable (Shor)	acceptable	7	7	soroban-env-host/src/cost_runner/cost_types/bls12_381.rs:6; soroban-env-host/src/cost_runner/cost_types/mod.rs:1; soroban-env-host/src/cost_runner/experimental/bls12_381.rs
BN254 (alt_bn128)		vulnerable (Shor)	acceptable	2	2	soroban-env-host/src/cost_runner/cost_types/bn254.rs:1; soroban-env-host/src/crypto/bn254.rs:4
ECDSA		vulnerable (Shor)	acceptable	6	6	soroban-env-host/src/cost_runner/cost_types/decode_ecdsa_curve256_sig.rs; soroban-env-host/src/cost_runner/cost_types/sec1_decode_point_uncompressed.rs:5; soroban-env-host/src/cost_runner/experimental/ecdsa_secp256r1_recover.rs:7
ECDSA	secp256k1	vulnerable (Shor)	acceptable	4	4	soroban-env-host/src/cost_runner/cost_types/recover_ecdsa_secp256k1_key.rs:7; soroban-env-host/src/cost_runner/experimental/ecdsa_secp256k1_verify.rs:6; soroban-env-host/src/cost_runner/util.rs:12
ECDSA	P-256	vulnerable (Shor)	acceptable	6	6	soroban-env-host/src/cost_runner/cost_types/sec1_decode_point_uncompressed.rs:5; soroban-env-host/src/cost_runner/cost_types/verify_ecdsa_secp256r1_sig.rs:5; soroban-env-host/src/cost_runner/experimental/ecdsa_secp256r1_recover.rs:7
Ed25519	Ed25519	vulnerable (Shor)	acceptable	13	13	soroban-env-host/src/builtin_contracts/stellar_asset_contract/metadata.rs:158; soroban-env-host/src/builtin_contracts/testutils.rs:7; soroban-env-host/src/cost_runner/cost_types/compute_ed25519_pubkey.rs:3

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

B. JSON

qsi.report/1 · SHA-256 9b85f3462552f82353205b89d9274711efee1e99748e272a6e7abd94e0172d1e

C. Input files

File	Size	SHA-256	Rec
.cargo/config.toml	6685	deaae172a164fb2ba5bfa0e6b3658fc8ef043a348c861936762a8a210c213ca0	0
Cargo.lock	66997	8004e3b6495cf7791937bff1131e216c8998def149562f66e2cf431ed99d3995	0
Cargo.toml	1714	d24c948dc1b2ed4c0bd856c41493867d866a4b4965b640ad6315db9716347152	0
rust-toolchain.toml	369	a93034bc4702710436624ddc42e89325be59ebc8e584e7c58b9b6c00fc2d3b2d	0
soroban-bench-utils/Cargo.toml	787	2020e9795c6dc4edfee39f36f1d78cf16b954f69ac16f9cba453485985d4edd3	0
soroban-bench-utils/src/lib.rs	75	4ed9af88c90c8137a175e9b2e583a3076ef13bc4911e133e889aa6f559caeeab	0
soroban-bench-utils/src/rusagev4.rs	2027	e775f766b27fcece33febfaf91e1b2e46b4ff0d7028a326909fd910dd75c7b17	0
soroban-bench-utils/src/tracker.rs	5923	71df7e2a505682276058fba51644bfe0f0d59934c327b9f4bb14ee6d59a6f8b4	0
soroban-builtin-sdk-macros/Cargo.toml	554	e2e52a67795c6540896ee63bfaf72c0b52ad318e78ca7028a1b02e961d8ff063	0
soroban-builtin-sdk-macros/src/derive_fn.rs	2815	627430878da696e94b872adc29d71390e866277ee7acf6c1b42e7cdd2ccf15a0	0
soroban-builtin-sdk-macros/src/derive_type.rs	8525	459416267127d646b9d105c16c046b210eeb22cd4c482511f268d8396120f215	0
soroban-builtin-sdk-macros/src/lib.rs	1734	7120a6c227f456234536bfef7c7f9d82918a8c89d8c25f3f16eaccc404a79aad	0
soroban-env-common/Cargo.toml	1521	07a820c9c891a409cbe763dfbdb2321d0103fa844ffc2bdefaffaed275a7f2be	0
soroban-env-common/build.rs	99	818b5844721186dd913412974faaff4bb45954e41b2cedeecb4b0a5ddcf90b35	0
soroban-env-common/src/arbitrary.rs	2047	aaebd099ab97097e1ac2482ac4c437c99736f75ca822cfdcec435ccf8b8fb662	0
soroban-env-common/src/bytes.rs	3066	768838872830e2352670e97ad9d9313c24db0efb8854847474739ed33c7c5735	0
soroban-env-common/src/compare.rs	7518	a61b9a80f87ee4b47ec8fb731fe2352ab4152941de87c0c3a4d4d032fa2f155e	0
soroban-env-common/src/convert.rs	19786	0ec5cb77ee4849bd270ce7ad629c793e921246f7f26e9c5d12bc6f412214b5ab	0
soroban-env-common/src/env.rs	18735	d2b81cd30de3099dc714503c6d87904b38d76e5da23a70c33dc9a36e9ead3150	0
soroban-env-common/src/error.rs	15600	efd9f516f411d55dd01a22e33f3418068d82c715290382e98a5cba4f58bdc46e	0
soroban-env-common/src/hash.rs	1501	15c9e63948d9ce9486581659f1aea95a08eb849416500a0a096f8d9d11ec1350	0
soroban-env-common/src/lib.rs	4018	57665754a7a74c76b2f6c90a6f846e45d917eec2b48427f7c606fb8049cf74d2	0
soroban-env-common/src/meta.rs	3279	4cd2287db41b82d68928def2b71ad815c1cf3e3ea5efafe8e4f02b9392001c31	0
soroban-env-common/src/num.rs	14398	9242b8a3eeb918cbc602ecbb6e185b3d8bc6cacf27bb50d35c7aad767de5ff0c	0
soroban-env-common/src/object.rs	6008	8241f6640f0fb0f0fa875fce49490d0fe8247c682a6eeee37391f2d93a4ec8a7	0
soroban-env-common/src/option.rs	907	870a61869cabc46e32f50c93605686f4d66f1952bf89f596b8fbfdcaf5deb38e	0
soroban-env-common/src/result.rs	1261	d3210db790113cf7950e76c15492358ca9a469159f8dee2c27c2dee91228c4b0	0

soroban-env-common/src/storage_type.rs	1474	056d61661c117c628f61dbc1307aa76a72254c69be680d9508f9e5598a977bc0	0
soroban-env-common/src/string.rs	2036	13ec3079a2f66e4affb3ecdeba15134f2941ef05076f544a195736dbd64bf8e9	0
soroban-env-common/src/symbol.rs	21593	8c11e956065de27f8d67d55eab1fdfacc515d31c57c3e8e30baa5d905d30d68d	0
soroban-env-common/src/tuple.rs	4295	fed0d299699f9ad2f429f68199534af4b319cfbc960dfdf488e9dc943fae95ad	0
soroban-env-common/src/val.rs	31320	c236111a028b7dcde91dfc1e816ed22f5eabe94b95499d03fccf089a73b6ef8e	0
soroban-env-common/src/vmcaller_env.rs	11627	ea4fba3d54f89bdaacb93ba6132955f6b9fb4d78e4d07890adee5cdb6f6da1d8	0
soroban-env-common/src/wrapper_macros.rs	12097	0228b5e35382d4dbca4b760f3f6a57d0a7ecbb9ffa897f16d445c82ce6e63e93	0
soroban-env-guest/Cargo.toml	625	65d7195e1ef69e1cba7c6aabd96072fe37482f0b1c7ecaeb17e17cc38b2ab937	0
soroban-env-guest/src/guest.rs	16376	d837794209e6f0cd20b4ee4de10dd2019417577fab6c6f4866da0397a6add69b	0
soroban-env-guest/src/lib.rs	561	b34cdc92dbf3880bd463d0c82b7df5866ac88d74b8bc42d8a9321eb9104dcfdf	0
soroban-env-host/Cargo.toml	5123	10866901b8a2f16b1876dd7e80ac5cb1c3b08339c6883f76a83c5cfa72a39e87	0
soroban-env-host/build.rs	283	f90c468c4ab2735a8edeca9f82ca31cf327a27fd8baf47f7417dab8653151005	0
soroban-env-host/src/auth.rs	129016	8307d3aeaf9c1a7c8199d2107714aff9c298ae299475ec4470ad28ead446e91b	0
soroban-env-host/src/bin/main.rs	1615	499367447384bdebd2654a72fe3f3f916e523762b2442e1b8417ea7d98d49e9f	0
soroban-env-host/src/budget.rs	64867	2ce3d7ba5c0c1ad1593a4e76c2ab1b7d88d5aede0501e875dd99902f8346fe80	0
soroban-env-host/src/budget/dimension.rs	8056	14d8a99a08a0844636e62d66a77e15d9a215929a65bc062777730f88ce532bfe	0
soroban-env-host/src/budget/limits.rs	4658	0080964f53ec49d2df64306354887cc5565b2628584e64548126a4b95e2eca1e	0
soroban-env-host/src/budget/model.rs	5668	70c47d818a27c286180e758280590839971b0cb071f227d33f96f2e935bcc3c7	0
soroban-env-host/src/budget/util.rs	7351	2e16740c47331fe2c88aad6bf3f80ede095f0259e5517cf3ff848a242495a6f9	0
soroban-env-host/src/budget/wasmi_helper.rs	5257	72c4869ce8a190b7afb82091692330700c6f43ba984164815d95ca3c1d62b6b3	0
soroban-env-host/src/builtin_contracts.rs	560	f0bd26290068aff0d567b8e950c3abd9545b9f6fbb50a521001e21f9b3a7ed1e	0
soroban-env-host/src/builtin_contracts/account_contract.rs	9242	7af518b93ec3124d9b38a67a8272e73a7f5af6a5f495e612d08fec87416aad3c	0
soroban-env-host/src/builtin_contracts/base_types.rs	14101	a66d54af8360d2c950f469f7aceb2a5ebb805e716b94b687dc15f366c0900f7d	0
soroban-env-host/src/builtin_contracts/common_types.rs	3474	e70a1ef6507838259fba577aefb9add282e7fe108ea4c98fa79d4b4af50f7348	0
soroban-env-host/src/builtin_contracts/contract_error.rs	1099	f763a1979d169c8df77ece0426b40a29686435b9b5da466243de60fd005761db	0

soroban-env-host/src/builtin_contracts/invoker_contract_auth.rs	6122	f3ae7c92ba3f2e9b80cfee73b1b6eba1b3edffbf2c7d92ee1ec649f618617690	0
soroban-env-host/src/builtin_contracts/stellar_asset_contract.rs	248	944142447b113d671f1e47dc8a2d8191ef4ff11167b793f2a4c2f4d72efc8e4f	0
soroban-env-host/src/builtin_contracts/stellar_asset_contract/admin.rs	704	ad70748dab976704bdf5fa87dfe30c0746280a24eb46c36a9c102aeb1f11980d	0
soroban-env-host/src/builtin_contracts/stellar_asset_contract/allowance.rs	5164	9bb65ada40aa4430359eb27a7e77aed2d90d7d7316e48b99dda71540eecb2fe3	0
soroban-env-host/src/builtin_contracts/stellar_asset_contract/asset_info.rs	3092	9a0583a6bec016f2822bc357baa2cce81c8e4ad3077fd8d5e2100172c0b22def	0
soroban-env-host/src/builtin_contracts/stellar_asset_contract/balance.rs	39345	671ea7ffb78b3c59b97ce417fac9cb5b4a6c8af5a592d8d51bd0b8a344108d29	0
soroban-env-host/src/builtin_contracts/stellar_asset_contract/contract.rs	14110	5fe6f53f7da37861501ae6e6205907f3473679a8cabe1eca61759d50210e997a	0
soroban-env-host/src/builtin_contracts/stellar_asset_contract/event.rs	4403	125cc56ff663883f629520b9ea1fc1978527c61b86867063cd57cba7730b211c	0
soroban-env-host/src/builtin_contracts/stellar_asset_contract/metadata.rs	6992	12ca5219bd19e98b81880a01e5fe99bfffaf1c5d283371bff34961763ac00331	0
soroban-env-host/src/builtin_contracts/stellar_asset_contract/public_types.rs	1318	84834c2fec3c9763ba73d89fdd44db86da9a35400e82401cd5ba04ed03874026	0
soroban-env-host/src/builtin_contracts/stellar_asset_contract/storage_types.rs	1113	655a0cec594b78db2eeccabfe03e27f2ae0537c65729e456b604cbbdb579f298	0
soroban-env-host/src/builtin_contracts/storage_utils.rs	362	216d16572291ac021fd3ee7a0ce6d4b1810f8dbee07fc2791bbf32236a75ed10	0
soroban-env-host/src/builtin_contracts/testutils.rs	14105	59edd2300fad57079e61b23aab7a44bcd62b22fcf16b525fbf7e01596e57c460	0
soroban-env-host/src/cost_runner/cost_types/bls12_381.rs	10204	965dff98b1d17f2c58dd07697781db1c8033193651fb14d4473b091fb6ab43c4	0
soroban-env-host/src/cost_runner/cost_types/bn254.rs	6857	b77f33938ce6e3aab58c9ce36abbe2d5ffe5f11643557264de93e493647711e1	0

soroban-env-host/src/cost_runner/cost_types/compute_ed25519_pubkey.rs	952	c679365a8825299630452cf28daf21f4e6cff1805ca06910bc4ce49300d04a7c	0
soroban-env-host/src/cost_runner/cost_types/compute_keccak256_hash.rs	916	c70f481dcea5eba30a4ae54ea476efec3fb3bf938d15987042aaeac222bbef3d	0
soroban-env-host/src/cost_runner/cost_types/compute_sha256_hash.rs	896	99eece10c8e2379a0d3675f618b6c9192a57ddd98bc0063cc886ab22d526b509	0
soroban-env-host/src/cost_runner/cost_types/decode_ecdsa_curve256_sig.rs	1407	6eae776ff6b8ee070b6d51f70a4b231bf78c2bd0e964c0cae695be06a66b251e	0
soroban-env-host/src/cost_runner/cost_types/host_mem_alloc.rs	951	53bd536e402308f09e0ead2a7825d8a7a88c6da46f0deb55ddce336d6a754a32	0
soroban-env-host/src/cost_runner/cost_types/host_mem_cmp.rs	957	db2b390ce0892f2b450a9707d8f3d54a9fe2438d88f73ae021263dc18b266a3c	0
soroban-env-host/src/cost_runner/cost_types/host_mem_cpy.rs	821	7ab8627489d5882ca2725d3e5872dc5cf3907890341065e5fa9149b121f11316	0
soroban-env-host/src/cost_runner/cost_types/invoke.rs	2183	451a7818021ac77658771411708fe709c328f8d29abb0906720cd4da4fbb069d	0
soroban-env-host/src/cost_runner/cost_types/ml_dsa.rs	5640	9b22af01136e5ea270aa291e845b33189d96bf106f893528a293e677cb9d9fc6	0
soroban-env-host/src/cost_runner/cost_types/mod.rs	1043	fd0538987c9bb5d51e38e84665ae18e2093e3b868f0f3d90efec7de08c2388f2	0
soroban-env-host/src/cost_runner/cost_types/num_ops.rs	1546	cf08647785a4ede49183258e305e4520b3db286536bf2a6e71583e76c89be32b	0
soroban-env-host/src/cost_runner/cost_types/prng.rs	1003	fac74f446f916302d6b27d59040ae6c72aa4decbfe766a862f9b0a2dceb75223	0
soroban-env-host/src/cost_runner/cost_types/recover_ecdsa_secp256k1_key.rs	1140	21256191032a43cd2584275eebae7e034db553ee6c90c16d6ba5e5e2392eb69a	0
soroban-env-host/src/cost_runner/cost_types/sec1_decode_point_uncompressed.rs	1155	06fda4cc5f09ca513953f45424b1aa5277fb1fa0805e598e63466ea6b45d1863	0
soroban-env-host/src/cost_runner/cost_types/val_deser.rs	800	19b4abd59908d36bc2c1a2d7e05fc67db21df215cf02bcf91ff42d760f18e6d0	0
soroban-env-host/src/cost_runner/cost_types/val_ser.rs	1385	2b3aa6b23feb1d99fdfc267146458eba577e8641124c8673a606fa0cb976b08e	0
soroban-env-host/src/cost_runner/cost_types/verify_ecdsa_secp256r1_sig.rs	1124	8b1e249924c8f329e563122f8e2c1d0e60fbc6c788de70c0dd4e9aeaf8f8b833	0
soroban-env-host/src/cost_runner/cost_types/verify_ed25519_sig.rs	1079	da86df368d3cdc769f57e48a6fd160ba5c7f1dab0a44876587875be1167215bc	0

soroban-env-host/src/cost_runner/cost_types/visit_object.rs	1056	c66e1ae7580a923564f8694f34d3ee9ff4c3de2b768529cf75a9f86e23c9c656	0
soroban-env-host/src/cost_runner/cost_types/vm_ops.rs	8989	3a7c74abaa9d8e628d3bc586d8b3056218a8a234bd8e4f35857e186cb4fcc135	0
soroban-env-host/src/cost_runner/cost_types/wasm_insn_exec.rs	9902	2c3bfc08db067c5d0320dfbbb55d477019e282d7c7a4d4c6c1c704b47a6b2983	0
soroban-env-host/src/cost_runner/experimental/bls12_381.rs	6378	154aed6d61b073065304ff1b68dd8eb3709a82108a7983d23a44cb01cb744ac1	0
soroban-env-host/src/cost_runner/experimental/decode_secp256r1_sig.rs	1472	2f7bf5f395b05ad7ae9ab37e404eae4c3bfa85d29eaa8d08becfcfef825b0a1e	0
soroban-env-host/src/cost_runner/experimental/ecdsa_secp256k1_verify.rs	1487	d5d2bdc6133fc64cdbe0765a2d5ee6069a9cfedb8b03943dff269770d30b88c6	0
soroban-env-host/src/cost_runner/experimental/ecdsa_secp256r1_recover.rs	1499	2e4920ce9df183016f9bf4765bde45f76b746f34d636b03d21647e8ee7bfb7f3	0
soroban-env-host/src/cost_runner/experimental/ed25519_scalar_mut.rs	1353	5b0755082fbcd387ee756a4fcb85b19abb7f632aabefddae79063463cd81b480	0
soroban-env-host/src/cost_runner/experimental/map_ops.rs	1005	5feb978d228ff705dcee86bb90cedaa328dfa196c6fdbbfc48468d6dfbd8cd74	0
soroban-env-host/src/cost_runner/experimental/mod.rs	2700	cc07f061fc69b5348c5cb1e8dbaaad46f4b06ff7a396245618d954e98d4c77cd	0
soroban-env-host/src/cost_runner/experimental/read_xdr.rs	1388	476f11215c5a4bdcebe94772688e45f28d96db4f5dfe4da029507cd01d4f3f3f	0
soroban-env-host/src/cost_runner/experimental/sec1_decode_point_compressed.rs	1756	299089e64c18ca9c4414a6461eaf023cbcfdbdca24db9f55edbf28f922771643	0
soroban-env-host/src/cost_runner/experimental/vec_ops.rs	1132	8116fc7fe3f8d9259a9a98a719f9a8a67d96e2159db1644fc1e14cf352aaae48	0
soroban-env-host/src/cost_runner/experimental/vm_ops.rs	2196	a253e366b149b7583c7e48b3aff4a982fca99e0194b526c5276fcaa81253323a	0
soroban-env-host/src/cost_runner/mod.rs	787	d8b4f3f225b31efed2489ac70a610d6a774514944bdbb5645431e9cfef8251e3	0
soroban-env-host/src/cost_runner/runner.rs	4203	98f216ef92f42ce6364d6aa13cffcc33a6da9fa344469ebe0b7093614ca71fc7	0
soroban-env-host/src/cost_runner/util.rs	4774	9cbbfe0c50bd607d1d879a9968fdd5ec69ae096faa34216a7bbbe7986e2bc57c	0
soroban-env-host/src/crypto/bls12_381.rs	31382	1d75840927326348ef2e75c4e0c8d40106187e942eae5aa6b8a767bd7f4083f2	0
soroban-env-host/src/crypto/bn254.rs	22741	91d5f195e217783d883732c31e1ce33c318c553663bfcd299805b88b61c94885	0

soroban-env-host/src/crypto/metered_scalar.rs	8459	b84d9c5e689fd4055ec2de080d3a3c84088943c49d1920bff296d9240f1f833e	0
soroban-env-host/src/crypto/ml_dsa.rs	9649	106302dab96ec9e9f99551b8b138a585b99bb86982fac5715ff37b99f18d4201	0
soroban-env-host/src/crypto/mod.rs	16449	cdfde68a22e3fc0acf01d44d0efd9024c1d46802bc1ebf88c1ca3bdac6994af9	0
soroban-env-host/src/crypto/poseidon/mod.rs	915	a4fc963a5c1761ea73004b2906789078d4eec801d10f63e11c76930a95742ae7	0
soroban-env-host/src/crypto/poseidon/poseidon.rs	4398	5846c5e04d6bf324f3acf731886de138f0b0885d56231ffd5894267a4979b13e	0
soroban-env-host/src/crypto/poseidon/poseidon2.rs	11235	7a56d57770d361a570eb485ee8631268ddeade991566f5383b8f99d715ab926e	0
soroban-env-host/src/crypto/poseidon/poseidon2_params.rs	4136	6af419d0ef7e91cea5a67c0a158fbe72111a50e3fd46ef9fad2be03528f57c48	0
soroban-env-host/src/crypto/poseidon/poseidon_params.rs	3957	32820b7efefe8f1e3260f5fd710e5e32f3f6cdc865952cf1584dfb638eb9c5bb	0
soroban-env-host/src/crypto/poseidon/utils.rs	1188	ad887f21baf85f2221d85818720c5b0705f1b15ca9e4dc52225e38b52685d158	0
soroban-env-host/src/e2e_invoke.rs	50421	3ae8d07e58e898ec8144c5127bc3435427d45591fd72c04ca6fe747f66ea0de4	0
soroban-env-host/src/e2e_testutils.rs	11749	27727b47a61951317a21b854bd3440dae8ee7ba95fd913f03a9d84a08c1ccef3	0
soroban-env-host/src/events/diagnostic.rs	5750	a66118d5747c50c89b1c7e1fa859b6b43b91f10add59353f274b115e3a171776	0
soroban-env-host/src/events/internal.rs	11050	44c6828241d86a89af619d88026b57b38a7915632a8d163716c4afe9ac6743be	0
soroban-env-host/src/events/mod.rs	10348	8e8c489af9b03c9783bc9e396e6bd3fbd9d88b26e53ef345c77756af344ab8fe	0
soroban-env-host/src/events/system_events.rs	1356	126b8da42733be7deea23b239008eeca261828c191862f26be1d759b82422f53	0
soroban-env-host/src/fees.rs	15773	08017d0c144556974f05c7511887c91bdf4f9650772bd1b7bb630667fb3be46b	0
soroban-env-host/src/host.rs	143192	29dab11703256fe8fa58cf1f76556540325a8ca55beec6f75b75539d0d74f44e	0
soroban-env-host/src/host/comparison.rs	30570	58acd6130f38aa371d8bb78c7780314a1c3afdc05ee6b8fbe9e0ca19f3003382	0
soroban-env-host/src/host/conversion.rs	34426	bccb03358f834c0558e7b077de23643c2a5cb2bbf13b427612c63c34a59a93f4	0
soroban-env-host/src/host/data_helper.rs	42914	24f431604013d735654eec72c0fec8e41ff4dd9b3e353dbc0a641450062fa64b	0
soroban-env-host/src/host/declared_size.rs	39159	d7dba05a3added5b195a2d3985fe7a6261055e30967cb71ccbedcafe54f91e5e	0
soroban-env-host/src/host/error.rs	17368	6d4fa87584f3dc60588f83ac2bc359f8b89f6ca2b89b6bbfc34a32f854155faa	0
soroban-env-host/src/host/frame.rs	59783	55fce1dfc160bdff39bb5ea7d9d57b8a7eb0edcc21d94b253544d7f0645e9221	0

soroban-env-host/src/host/invocation_metering.rs	82843	ae1340afaadfed817c60897f244ce60494b38783e461f6d587a4ae55b21485fa	7
soroban-env-host/src/host/ledger_entry.rs	5565	101f183dfa65ec07d52bb1a96f3656de5121a301bcc96000fd058cefa5b39fbf	0
soroban-env-host/src/host/ledger_info_helper.rs	1502	9144a723b7119e868776cb21260b2f44ce286bc6a6485ada29c6133fc71e33ae	0
soroban-env-host/src/host/lifecycle.rs	26854	e842de9f63a3d3f833d460d52914a361026dfd823dc51c5a82ca46fc5a897ee1	0
soroban-env-host/src/host/mem_helper.rs	18356	7572dd84a13b22d152005ea55d3ee37ce2293da6d013382114815b8b9ec1e58e	0
soroban-env-host/src/host/metered_clone.rs	27328	e987eab2736380414d73d6d353743c42410751fac027630a9172fd6e812c1469	0
soroban-env-host/src/host/metered_hash.rs	2482	3792f8eea8df5b983fb363b0a282f8e91e3b7dfca5a424484cb5d3e0ae5c056b	0
soroban-env-host/src/host/metered_map.rs	15359	f81c4b76b8487f4f6d849751cd26353bf8778ecf0916266cf2e5fb09fd2e5b5c	0
soroban-env-host/src/host/metered_vector.rs	15101	01cb08e2b689a539d06d23518fb7f6507be48677e3b30919cd28151d8dd78da5	0
soroban-env-host/src/host/metered_xdr.rs	2650	22e70db0f290b025fb1e992b230564210a7b315c9ea7dca1e3e28609c64b7a52	0
soroban-env-host/src/host/num.rs	3676	db5c486082fc7c58daba77fb9690801c6e68601b3db7a6a2de9ed04590327d3a	0
soroban-env-host/src/host/prng.rs	8157	eeed1bff44661301241a4bd31c87705cc202eb8efd829763646947091bb6de39	0
soroban-env-host/src/host/trace.rs	13358	6de79ddfdcc146c188a6f340329d9ec62dfe7cfea46a534b3d8734e487855226	0
soroban-env-host/src/host/trace/fmt.rs	8072	0436f0f15540985f007b728d426470218ea463ed8b1dde23dedee0c08b986b07	0
soroban-env-host/src/host/validity.rs	1818	b5c6baa004d08a398e6553f5336c5c1dc2ccb099dfdc0ed708ed967147c883d5	0
soroban-env-host/src/host_object.rs	22259	f48df34cfd08f6a06c2b6a7e34e227524832648db1999792ff3f656d308e916e	0
soroban-env-host/src/ledger_info.rs	1348	5a7d202b945c0e639624746f926040eb873f640217652d55fb68fbc5acfb1f4c	0
soroban-env-host/src/lib.rs	2295	066328815dd01b7616d29bf56f74cf089c53cac95ae66910b44a56befe931cbb	0
soroban-env-host/src/macros.rs	3069	40406f573892430b2bdfce358923e0cd47bd06e0c2ab7d472a15f09b9a06b57d	0
soroban-env-host/src/storage.rs	39421	03831101eefc0fa9957ec690152f932aec42b3f9b7282e4a60da813dd756e3aa	0
soroban-env-host/src/testutils.rs	47592	b39bf85a8500949900f290c9d934ac29540d7f0a9ffa42e4a5af024b4566d78f	0
soroban-env-host/src/vm.rs	20023	dcaa075b50bb4aa900ef1dec78a715fcdcb2e189f80d61a85cc03aa9d5676ae0	0
soroban-env-host/src/vm/dispatch.rs	13945	6814aed8a87857b728a4a0dcffd959df516bad781875fe874ecc035e232e0eae	0
soroban-env-host/src/vm/fuel_refillable.rs	2310	b8e65b7971534028a2e4f2e865653cee23252a4b750ba5c9da35071b7b1cb110	0
soroban-env-host/src/vm/func_info.rs	6266	80fa97bf3e4b23155b0d52e9694d3f3351f2109f61b7a2bdf1df591f4df0d579	0

soroban-env-host/src/vm/module_cache.rs	7959	240ce8aa32e6723bde86891ba02930d7dc723ffee8af1411c9ae9262ed1fcf0e	0
soroban-env-host/src/vm/parsed_module.rs	31859	67900843e0903d307bbd93696950cc9943a79be6b4e55fd7a9a2fb0795e048ce	0
soroban-env-macros/Cargo.toml	710	ceca1d0ef5d050e4122b778a6f922ff9638365556a49398cf57192a0029bb858	0
soroban-env-macros/src/call_macro_with_all_host_functions.rs	5468	c09e407ee43ff22459be37984b927fee60e24fc649950d0c55f349030a11e52c	0
soroban-env-macros/src/lib.rs	7805	363a909ad723b7b97b568793f865979d935bebe102b055f9ec88daca1bdf2044	0
soroban-env-macros/src/path.rs	561	6d186fe3120e11b1deb22d5faa6005b68ded100c76cfde483d729fe8ea15c813	0
soroban-env-macros/src/synth_wasm_expr_type.rs	12311	350dc56852495c24af799b1398a490f31eba313fc273a106a6997c346b96ed0f	0
soroban-fuzz-targets/Cargo.toml	689	d30c34a4f158a2ccc57a4a4b9f4437edcd1d42e4cb6bb6e7aabc5e9de08bc906	0
soroban-fuzz-targets/src/expr.rs	6631	a73a323a5cbdb1bc9e48f233acb5098941c4d786b8882c312fdc0d414e604534	0
soroban-fuzz-targets/src/lib.rs	1150	3515e355cf1166e6786fb8bb3de69fad701890bd02e94272eaf08bb05d34eebf	0
soroban-fuzz-targets/src/wasmi.rs	4166	7466fbb5786c8e457ea19c947f1cff7a74b21ee93a477606b082179c21447115	0
soroban-simulation/Cargo.toml	1049	f12c43a7bfbb5def7e8a3d42c3e6f55f7330b10394367ac789eaef38fa50f58b	0
soroban-simulation/src/lib.rs	253	c554a9c4dc9655d33e65137f3e6b8f1335b256da59ea5460150c8b4e37b3b752	0
soroban-simulation/src/network_config.rs	7233	4f94adec79e4ec265a9b3760de3e2e5e39ce08ccf4ed380406f51e6b31ac1bec	0
soroban-simulation/src/resources.rs	12456	35702ace20e4c6712bb4680ab0b6f651749e8cb56367629e2e3daae07718ba03	0
soroban-simulation/src/simulation.rs	15415	7278d70854bd6bed517560045aa282014568533401e5965b8d71813c5f3f40b7	0
soroban-simulation/src/snapshot_source.rs	9727	bb6febacc3e5729108305606c7724931ccc7ef67db9806f2b17f0097810e387c	0
soroban-simulation/src/testutils.rs	2760	030d98797f9bbffee2409a0166d649042158a764c7982be9fc9f698f39b336f7	0
soroban-synth-wasm/Cargo.toml	809	4115f43e6325b4964960785e38559d7295734d4eac2ab7cec49fc33bac039ac5	0
soroban-synth-wasm/src/expr.rs	3370	20abb89282b4fcc30d85aa162b3edf8c49cd59a5d90ebc6bfd6ae6a24eacc3cd	0
soroban-synth-wasm/src/func_emitter.rs	15938	38d2d41444363114390bd9272701b0441a27402584df4983a08e075b7b9440e2	0
soroban-synth-wasm/src/host_funcs.rs	3306	5e70c2dfd0919f3f262b88e19aaf38b660cde443636533fb8c44bcfc08ba18f5	0
soroban-synth-wasm/src/lib.rs	1130	b94fdbd1d86e3438c269123a6f6c9426e0791bf3b60d81d7b0b40dceb9544328	0
soroban-synth-wasm/src/mod_emitter.rs	16360	cba054710ff6760889e36d128beeca2f4f94fbf1abacd82094daedafddbf0eec	0
soroban-test-wasms/Cargo.toml	241	6eef4c0875ef4d266d255a5f5c6bdfee377c1d347b4f7e5418f6ee1134739280	0
soroban-test-wasms/src/lib.rs	8515	eba4444d5652375364351b7ebf6cfe43575ac59506d10f5a6abb3c28f08d04b0	0
soroban-test-wasms/wasm-workspace/Cargo.lock	45193	d8e0f8458198b54e0361a56f9d1184b6ad0251bab768e6a88441953261c36b03	0

soroban-test-wasms/wasm-workspace/Cargo.toml	2761	8458693e1750830a592cf3594bc2d24c39d312b8863bdddcf3eb53e53725d6e5	0
soroban-test-wasms/wasm-workspace/add_f32/Cargo.toml	309	549e4c2516c26da535fbc0adede29e1e326c418737f90a536b6edb4059bd3eb2	0
soroban-test-wasms/wasm-workspace/add_f32/src/lib.rs	336	099703bd2f3f4088add74ae0d0f900d1cddb16967d483ca76480bd37e5aaa525	0
soroban-test-wasms/wasm-workspace/add_i32/Cargo.toml	309	dd8f899b62614dd9aa7b485eb5de3b02b65c0f341c828e7c86fa01c0ef893d9f	0
soroban-test-wasms/wasm-workspace/add_i32/src/lib.rs	275	609f7db6583e31743bb9b6baa6d7f9f272b3d0f76cc9bc633244bdadc1ae7856	0
soroban-test-wasms/wasm-workspace/alloc/Cargo.toml	329	01be473ad078fed7227bf22ed802501063bdc72f25c1e4a53c69972e9c68ee18	0
soroban-test-wasms/wasm-workspace/alloc/src/lib.rs	708	3ad94be5c06b55ed4639bb79f59be677ae9f8c38884bccb249bade8e7f4ad870	0
soroban-test-wasms/wasm-workspace/auth/Cargo.toml	312	f04a346b02d6b304d0b1fe2f724f742545a48ed26bce353fc75d868d47201ba1	0
soroban-test-wasms/wasm-workspace/auth/src/lib.rs	8544	2e0064ca991f83c3a00177f052bbc5c053087546ed1e42a944c7af810bb03621	0
soroban-test-wasms/wasm-workspace/builtin_auth_delegation/Cargo.toml	306	585673f879ec8063f7cc5b6a7e5e714ab5315f3a3fa2f054d13e8e65b9b9bc24	0
soroban-test-wasms/wasm-workspace/builtin_auth_delegation/src/lib.rs	4959	aea2b21df2e94d204ff31f7ccdea5efdcc044afce347060732dd47581482ac1a	0
soroban-test-wasms/wasm-workspace/check_auth_instance_storage/Cargo.toml	310	5d3baa922a48fbb24bf0fd4d6caf9057a6f0f4f0de307fdbf4ade6dd119b5862	0
soroban-test-wasms/wasm-workspace/check_auth_instance_storage/src/lib.rs	1900	8aaca638c110973cec3b5aa5fe3e45b77ed12f9e189120c88c0a7ba3737fb92d	0
soroban-test-wasms/wasm-workspace/complex/Cargo.toml	293	0067ed9e3a4473fa625554efc2b026ff83ab9211a7ae243dbdb6afebb6fb9245	0
soroban-test-wasms/wasm-workspace/complex/src/lib.rs	1522	7fb764db8fd1ff735f0077daa0c0f8fbca952fb621075820dd741321a27b37d0	0
soroban-test-wasms/wasm-workspace/conditional_account/Cargo.toml	302	9dec27ee4ce2269cb4fcf2b36bf97314ca3e666314f8361e3f352b86e184cc8a	0
soroban-test-wasms/wasm-workspace/conditional_account/src/lib.rs	1190	8373ab257590bdc0023784bdcd8c465630ca631a1f820b1db1e6fb9b88fbf4ab	0

soroban-test-wasms/wasm-workspace/constructor/Cargo.toml	294	ff1e3d8dc9d364c3fe88c5ebd67a52db55b36c761b5bb0827cba0a25ce5bce18	0
soroban-test-wasms/wasm-workspace/constructor/src/lib.rs	1058	c423ed6d892c0286acc21a9fb78cb7bb159b247dce09c5c9b01b29cdcc246d76	0
soroban-test-wasms/wasm-workspace/constructor_with_result/Cargo.toml	306	2e96ef48cc040386131b56ed8a1d555850283e07eaf5087ef8d62abdb5452900	0
soroban-test-wasms/wasm-workspace/constructor_with_result/src/lib.rs	621	32a1d47c7d3c10309b7aba51b7ac5138c80adcbed19b0726950c05b8946cf72d	0
soroban-test-wasms/wasm-workspace/constructor_with_return_value/Cargo.toml	312	59d0b78167d61493e8e991da525ba1e65976a14b7c9ea8b06ad944c19b9b8798	0
soroban-test-wasms/wasm-workspace/constructor_with_return_value/src/lib.rs	234	c30ae802d69000e032c778d0558e7a3cd1fc93750e69f617a27d5a281b812522	0
soroban-test-wasms/wasm-workspace/contract_data/Cargo.toml	299	77083980ef46d4854eb10c3fc40df19e910dcac6df40935a6af45b0eb3d17332	0
soroban-test-wasms/wasm-workspace/contract_data/src/lib.rs	3353	dbb18e27f6dff3765423e305add70da6a038f1a81835d0479b56c2ecce7733f1	0
soroban-test-wasms/wasm-workspace/contract_sac_transfer/Cargo.toml	299	9e6a1c69f818e4c6a52f70f0340235131b9af4b81faeffed68f4ded27154d277	0
soroban-test-wasms/wasm-workspace/contract_sac_transfer/src/lib.rs	1284	bbfce0bd50690fb25cce60b10178aec5d0292b1f3423cd59013297764bbd35d8	0
soroban-test-wasms/wasm-workspace/create_contract/Cargo.toml	317	281d6a213ecfeb71d508b60fb9240834123a0c7fb9b83d6690a7ef78625539bb	0
soroban-test-wasms/wasm-workspace/create_contract/src/lib.rs	419	d999ae4431e2cca2ff8500a042c58581208a70775eada7264e8985de95b11138	0
soroban-test-wasms/wasm-workspace/custom_account_context/Cargo.toml	305	92af459b76ad60e3ce347fcdc67aa5a1ad051c49b4adb77f7c67a55589badc64	0
soroban-test-wasms/wasm-workspace/custom_account_context/src/lib.rs	601	b657abdf54831ee754ff23623c5de4d0723013371aa201838112bc49df3bfb45	0
soroban-test-wasms/wasm-workspace/delegated_account/Cargo.toml	300	7908ae88dab514ec6d70894655953384c7da38a7d4f969c6d63ede8e23c6a439	0
soroban-test-wasms/wasm-workspace/delegated_account/src/lib.rs	868	36ad27bafba63f3f84003fc542a5ea676ae40bd5cf95d76532fcbf15f57c64e2	0

soroban-test-wasms/wasm-workspace/deployer/Cargo.toml	291	0a745dea42f11644e128a60ea95b230fc9228d657d1e59bd0505c06e8621d4d8	0
soroban-test-wasms/wasm-workspace/deployer/src/lib.rs	1873	ade03dfecc57c2a865c94f0e96203aae80eb6b5a081827a7247768a55fd561cc	0
soroban-test-wasms/wasm-workspace/deployer_with_constructor/Cargo.toml	308	7cd7dc3819357f91f599199090fd0c56e5ffa2f8e05e631255fb4e22330adce6	0
soroban-test-wasms/wasm-workspace/deployer_with_constructor/src/lib.rs	5011	7648828e97d74ed7ece814fa105f4ad2bbff85acc1e599b62e3d5923a24aea54	0
soroban-test-wasms/wasm-workspace/err/Cargo.toml	347	49a7e63a73faaa259430d07f3f4bfc4248053afd571a89ddad82b8b488e5c8be	0
soroban-test-wasms/wasm-workspace/err/src/lib.rs	2881	6d421da68d5bcfb4dec07757387e48833cb5f3bbe6ce99dfd8730a08605aed23	0
soroban-test-wasms/wasm-workspace/fannkuch/Cargo.toml	310	b6ec9f13152db30ec92fb459ede39594cbc77404f8e176e06baa71b323a11ef0	0
soroban-test-wasms/wasm-workspace/fannkuch/src/lib.rs	2236	f8af8a42fd6fc33fde15a97091e4382bba5c851e44210cf3a840f70d8ef81c97	0
soroban-test-wasms/wasm-workspace/fib/Cargo.toml	305	30b7bf6023952b31b3a9d8cb1df2aa7223149253f8678452726b0d0df58035be	0
soroban-test-wasms/wasm-workspace/fib/src/lib.rs	268	c957875f4c1b35d053c3b2d746bcf748667913f12e33be6e03d244b537521d84	0
soroban-test-wasms/wasm-workspace/hostile/Cargo.toml	350	a828d8f58a569be29dc2f6c64a35953cf15829e6ad1925b69d76d2ff08709752	0
soroban-test-wasms/wasm-workspace/hostile/src/lib.rs	3124	c2eedf2f69c9f8cefbcf0e12967b76413d1b42e347f6478c221e8b294c6227f9	0
soroban-test-wasms/wasm-workspace/hostile_large_val/Cargo.toml	311	1317a69cd05283c9f027d3211fb3b37d9ae9757257a5565ae5b145be0680c8eb	0
soroban-test-wasms/wasm-workspace/hostile_large_val/src/lib.rs	2553	dbf1d00afd9f5f62c0bab4a4e51114fb876585262f8da8af1f030f04031d9a02	0
soroban-test-wasms/wasm-workspace/invoke_contract/Cargo.toml	317	c1f763f78de7b09d9095efac80104d088f6bd41ecc2bd6fb59e345e90c5beb67	0
soroban-test-wasms/wasm-workspace/invoke_contract/src/lib.rs	1943	331251706eeb7ca4c8f64877b2f4e2176b7195299989a6271db21db3c0a70357	0
soroban-test-wasms/wasm-workspace/invoker_auth/Cargo.toml	320	cdcf51951afe17159a5b6297b81e29ecbb785db2b3a654f7378a70a7f62fc726	0
soroban-test-wasms/wasm-workspace/invoker_auth/src/lib.rs	5921	6deed63acc70a7cf0c92f307243454a2ba4b1d2aed10cbafb59ed07207d64b64	0

soroban-test-wasms/wasm-workspace/linear_memory/Cargo.toml	341	907403fc34c319e3f0c99cc42b49584c108d39a846eca72924ac03a49d6a707c	0
soroban-test-wasms/wasm-workspace/linear_memory/src/lib.rs	2861	998f890f085a31c269dee25aa5ac3eef9a62efc4166d2d26377073eab3574a3d	0
soroban-test-wasms/wasm-workspace/loadgen/Cargo.toml	293	54a775e24e6a6b93bd1a91e8a8f67c9ba8a34893ef5dc66b70aa2e4b8c3082b8	0
soroban-test-wasms/wasm-workspace/loadgen/src/lib.rs	2035	9a2077b1af173b0dad443032321f21dce3b6966d0ae88ff2c0043ec6f3859640	0
soroban-test-wasms/wasm-workspace/no_arg_constructor/Cargo.toml	301	cdd661d64e2e9731bdbc49637372195660a1f21304f516af3ec7fc6164fec8c2	0
soroban-test-wasms/wasm-workspace/no_arg_constructor/src/lib.rs	696	f49c7035942d9351dde370ed4ea4e00698f76f66f2cee13079ccc0636acdc413	0
soroban-test-wasms/wasm-workspace/recursive_account/Cargo.toml	287	ba3460e51138adcf00bcd2fe7e59d6dc01a78fa752ead699befae8c3c607ba16	0
soroban-test-wasms/wasm-workspace/recursive_account/src/lib.rs	1819	99df3ded80c336870ed36935ca34af8306c1644bcc127f8adee9e6cb26ff2a24	0
soroban-test-wasms/wasm-workspace/sac_reentry_account/Cargo.toml	297	a8627c01b2129d0dae15ed2f709724cbebcb641ab9f5c66b9893b57e9c7078f4	0
soroban-test-wasms/wasm-workspace/sac_reentry_account/src/lib.rs	1176	44c4a73769a6db79991b2b981560e4c62df6a07a5a467550c8995f5333af2164	0
soroban-test-wasms/wasm-workspace/simple_account/Cargo.toml	300	233daf8057939469264afc3dd17b237b7ef423085d4eea909e79ddf5a41b2cbb	0
soroban-test-wasms/wasm-workspace/simple_account/src/lib.rs	948	7c74cf48ea7db5ed1ff209926ba7d98ba4c32127d9e52e229a3c90eedfed4e28	0
soroban-test-wasms/wasm-workspace/sum_i32/Cargo.toml	309	c37d0a5d2621bb8edb11aaa0a111d4c0d8afa4e1a65df8dfc404136711141d25	0
soroban-test-wasms/wasm-workspace/sum_i32/src/lib.rs	561	4b1c69bf2e59646c6202ce0e9e460a7d22eea3490e517ecc4f56cfc8e4b8f49b	0
soroban-test-wasms/wasm-workspace/try_call_sac/Cargo.toml	295	a8ef3fd8c92173dbc8e0975d0ba005063a495e49bd2be96fec39d73c5c057599	0
soroban-test-wasms/wasm-workspace/try_call_sac/src/lib.rs	435	e469644aac889f9613c96038630e05302321aede066868c0ac67e55f68508c5f	0
soroban-test-wasms/wasm-workspace/update/Cargo.toml	321	add583b889fe9dba5aa81ac0e5aa0a4d1cf11df49c0600e720b71b6c932fc1c2	0

soroban-test-wasms/wasm-workspace/update/src/lib.rs	945	d0ae9d5190e0f6251ee7182be9c052e068695ed305c7666caf4b455789c72c93	0
soroban-test-wasms/wasm-workspace/update_exec_ref/Cargo.toml	314	9422d19fe2ef02653a2083bfca58d6ae52eddf7a04071a96c66cd327501a0705	0
soroban-test-wasms/wasm-workspace/update_exec_ref/src/lib.rs	1105	064e0dcae88d9e531fcf0d64240f1d650ab7c77afeaac0d07bec787a95dbc8a5	0
soroban-test-wasms/wasm-workspace/upload_contract/Cargo.toml	317	521c2d07450d0ca5174ed43df2b01b0e85341b9291813f03476a8aec1db2472d	0
soroban-test-wasms/wasm-workspace/upload_contract/src/lib.rs	252	5033bb1c628243bdaecc55b1c7676b20b22f73fe4842905b5bb92a1a111061c9	0
soroban-test-wasms/wasm-workspace/vec/Cargo.toml	289	8e78cf5495f3569dcb16e392caec5420e0c4e38682a79bee9f24bdf1bd2ef8e3	0
soroban-test-wasms/wasm-workspace/vec/src/lib.rs	346	a358381709762177476df8c9dcacdc6406e499b81de7523278f3fbf5212aa7d3	0
soroban-test-wasms/wasm-workspace/write_upgrade_bytes/Cargo.toml	330	6f5ed7b39a290835d688ae0aa0ae465b065d7b02fc6f758f08e3df5fc090d95c	0
soroban-test-wasms/wasm-workspace/write_upgrade_bytes/src/lib.rs	1003	8a0cc6e5f77713ee726982d933260e90732e70873c0a2bb8cc662ec651472c6d	0